

Propositional Structure and Higher-Order Metaphysics

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1 Introduction

Higher-order logic is a generalization of first-order logic which allows quantification into more grammatical positions than just term position. This higher-order quantification is meant to capture structurally similar inferences to those found in first-order logic, such as existential generalization. For example, in first-order logic we may make an inference from “Audrey is brave” to “Something is brave”. In a higher-order logic which allows quantification over predicate position, we may also make the inference to “Audrey is something”, where ‘something’ is to be understood as an existentially quantified variable in predicate position. For clarity, we can express the formalized versions of these two inferred sentences as $\exists x(Bx)$ and $\exists X(Xa)$, respectively. But while the structural similarities between these two sentences are immediate, it is not as clear what the second sentence actually *means*. What should we understand ourselves as doing when we quantify into predicate position? We will come back to this question at the end of the next chapter.

What is clear about higher-order logic is that it allows us much more flexibility in regimenting the more unwieldy parts of natural language. This point was recognized by Richard Montague in the 1960s and led to his three groundbreaking papers “Universal Grammar”, “English As a Formal Language”, and “The Proper Treatment of Quantification in Ordinary English” which form the theoretical core of contemporary work in the semantics of natural language. At the time, it was widely believed by both logicians as well as the emerging linguists led by Chomsky that Tarski’s denotational semantics could not be generalized from the artificial languages of logic to the complicated languages humans naturally speak. Montague’s insight was that the increased expressiveness of higher-order logic provided the missing bridge between natural and formal languages, and since then higher-order logic has been an integral part of research in the philosophy of language and formal linguistics.

Yet despite higher-order logic’s widespread success in philosophical and scientific inquiries into language, its uptake by metaphysicians has been more cautious. Without a doubt, Quine’s famous arguments against the coherence of higher-order quantification and the failure

of higher-order logics to count as logic proper caused many metaphysicians to stay strictly within the first-order realm while theorizing about modality and ontology. However, this picture has changed drastically over the last 25 years. Heralded by Tim Williamson’s *Modal Logic as Metaphysics* and Cian Dorr’s “To Be F is to Be G”, higher-order metaphysics is a research program which takes higher-order logic as an indispensable tool for understanding the nature of reality, not just the languages we use to talk about it.

What most proponents of higher-order metaphysics take to be the greatest strength of the program is its ability to shine new light on old, recalcitrant debates in metaphysics. Williamson (2013) uses a higher-order modal language to argue that the stagnant debate between possibilists and actualists is really the result of confusion and imprecise language. If we instead adopt higher-order tools, Williamson argues, we are able to make substantive progress on foundational questions in modal metaphysics. Dorr (2016)’s goal is to use higher-order languages to understand *identifications*, claims of the form ‘to be F is to be G’. While first-order approaches to these questions treat them as questions about the identity of properties, Dorr’s higher-order approach allows for a deflationary reading of higher-order identifications and thus room to explore their nature in full generality.

Of particular interest to higher-order metaphysicians are identification questions of propositions. First-order approaches to the identifications of propositions have often favored what is called a *structured* view. Proponents of structured propositions argue that propositions are complexes which have internal structure. It then follows that identification of propositions supervenes on the identification of their constituent parts. However, many proponents of higher-order metaphysics argue that the tools of higher-order logic show that the structured view is not just implausible but outright false due to what is now called the Russell-Myhill paradox. The paradox shows that, under the standard formalization of higher-order logic, adding a structured theory of propositions contradicts Cantor’s theorem and hence renders the logic inconsistent. So, the proponent of higher-order metaphysics claims, we once again have a case of higher-order resources giving a clear answer to an old metaphysical debate.

Or so it seems.

I do not believe that the prospects for a higher-order theory of structured propositions are as dismal as the Russell-Myhill paradox suggests. In particular, there are changes we may make to the underlying higher-order logic which block the derivation of the inconsistency. Such changes would fail to be convincing if they were designed in an *ad-hoc* manner just to defuse the paradox, but I believe that these changes have independent motivation which cohere with the general project of higher-order metaphysics. Moreover, I believe that opponents of structure fail to make a necessary distinction between the relevant logical principles and the general metaphysical principles involved in the project of higher-order metaphysics. Drawing out the contours of this distinction reveals the non-arbitrariness of changing the underlying logic.

The rest of this thesis will proceed as follows. In Chapter 2 I introduce the higher-order logic we will use throughout the paper and present the standard way of thinking about it according to higher-order metaphysics. This will give us our answer to the question of what higher-order quantification means. In Chapter 3, I show how higher-order logic gives us a particularly elegant way of pursuing the identification question for propositions and give an initial argument for a structured view. The chapter then finishes by deriving the Russell-Myhill paradox and isolating which elements of our logic are in tension with the structured view. In Chapter 4 I present and defend against arguments opponents of structure have given, namely that the structured view is in tension with widely accepted principles in metaphysics and that its motivation derives from conflating worldly metaphysics with representational doxas. In Chapter 5 I offer arguments against the parts of our logic which conflict with structure and offer a revision to the logic which I believe better respects the topic-neutrality which is constitutive of logic. Finally, Chapter 6 concludes with a brief summary of how far we've come, and outlines the two main takeaways I believe my work has, one local to the project of higher-order metaphysics and one large about philosophical methodology.

2 Higher-Order Metaphysics

In this section we will first present our higher-order logic in detail. We then move on to discuss the philosophical interpretation of this logic which guides the project of higher-order metaphysics.

Before we begin, it is worth outlining the differences between a formal *language*, a formal *logic*, and a formal *theory*, since this distinction will help make clear what we are doing in this chapter and what is at stake in later chapters. A formal language is a collection of formulas—strings of symbols—which are formed according to a fixed set of grammatical rules. The language is the set of formulas which are well-formed according to these rules. A formal logic is a subset of the language: in essence, a formal logic is the particular formulas of the language which are taken to be the “logical truths”, and nothing more. What determines when a statement in a language is a logical truth is a core problem in the philosophy of logic, but the standard answer is “true no matter how we interpret its non-logical symbols” and this is made precise by the familiar Tarskian semantic theory of truth. A formal theory is also a subset of the language, but it contains “non-logical” truths in addition to the logical truths. These are truths which follow only once we fix some particular intended interpretation of the non-logical symbols of the language. A formal theory, then, is built on top of a logic and is an attempt to capture the truths of some particular domain, such as mathematics, science, epistemology, or metaphysics, as opposed to just the general, universal truths of logic.

The presentation of a logic usually proceeds in three steps: syntax, semantics, and then proofs. The syntax of a logic tells us how to form the grammatically correct statements of its language and the semantics tells us what these statements mean. The proof system unites the syntax and semantics by capturing the structure of the semantic entailment relations between the statements via a collection of syntactical inference rules.

Our presentation deviates from the above strategy in that we present our semantics only after a rigorous account of the syntax and proof system. Moreover, the semantics we will consider is not a formal semantics on the same level of rigor as the syntax and proof

system. While there is a rich literature on the model theoretic semantics of higher-order logic, those sorts of interpretations are not relevant for our purposes and we will instead focus on elucidating the nature of the intended, informal interpretation of our language since this interpretation is the one central to the higher-order metaphysics project.

2.1 The Higher-Order Language $\mathcal{L}$

The language of higher-order logic is at its core an extension of the language of first-order logic which allows quantification into all of the grammatical positions of the language. Yet despite this simple characterization, there are myriad ways of formalizing higher-order languages.¹ The formalization we will consider is that of a *functionally typed lambda calculus*. This is to say that we regiment the higher-order parts of our language through the use of *types* and *λ -abstractions*. We will refer to the higher-order language we are defining as $\mathcal{L}$, and we refer to expressions of the language as *terms*.²

Types are strings of symbols which correspond to the grammatical categories of $\mathcal{L}$. The set of types of $\mathcal{L}$ is given by the following definition.

- e is a type of $\mathcal{L}$.
- t is a type of $\mathcal{L}$.
- If σ and τ are types of $\mathcal{L}$, then $(\sigma \rightarrow \tau)$ is a type of $\mathcal{L}$.
- Nothing else is a type of $\mathcal{L}$.

The symbols e and t are themselves types of our language. We refer to e as “the type of individuals”. We may think of terms of type e as playing the same syntactic role in $\mathcal{L}$ as names, descriptions, and more general determiner phrases do in natural language. We refer to t as “the type of propositions”. Terms of type t make up the formulas of $\mathcal{L}$. Types e and

¹For an encyclopedic overview, including of the strategy we will employ here, see Bacon (2024).

²As is usually the case in formal logic, we may make some small conflation between use and mention for the sake of ease of presentation. When the distinction is philosophically salient, more care will be taken to practice good quotational hygiene.

t contrasts with σ and τ in that the latter two are not themselves types in our language. Instead, they are meta-language variables we use to speak about arbitrary types in $\mathcal{L}$. Types of the form $(\sigma \rightarrow \tau)$ are referred to as “predicate types”. We adopt the conventions that parentheses of predicate types are right-associative and outer parentheses may be dropped. This means that, for example, by $\sigma \rightarrow \tau \rightarrow \rho$ we mean $(\sigma \rightarrow (\tau \rightarrow \rho))$.

We will now use our set of types to give the definition of the terms of $\mathcal{L}$. First, terms of $\mathcal{L}$ come in two kinds: simple and complex. Simple terms are either constants or variables of some fixed type. The set of simple terms of $\mathcal{L}$ are given by the following definition.

- For every type σ , each of the infinite variables $x_1^\sigma, x_2^\sigma, \dots$ is a simple term of $\mathcal{L}$.
- For every type σ , each of the infinite constants $c_1^\sigma, c_2^\sigma, \dots$ is a simple term of $\mathcal{L}$.
- The constants $\neg$ and $\Box$ of type $t \rightarrow t$ are simple terms of $\mathcal{L}$.
- The constants $\wedge, \vee, \supset, \text{ and } \leftrightarrow$ of type $t \rightarrow t \rightarrow t$ are simple terms of $\mathcal{L}$.
- For every type σ , the constant $=_\sigma$ of type $\sigma \rightarrow \sigma \rightarrow t$ is a simple term of $\mathcal{L}$.
- For every type σ , the constants $\forall_\sigma$ and $\exists_\sigma$ of type $(\sigma \rightarrow t) \rightarrow t$ are simple terms of $\mathcal{L}$.
- Nothing else is a simple term of $\mathcal{L}$.

We adopt a handful of conventions regarding the simple terms of $\mathcal{L}$ which are similar to those adopted for standard presentations of first-order logic. For instance, we often use $x, y, z, \dots$ and $a, b, c, \dots$ for variables and constants of type e respectively. We also use $p, q, r, s, \dots$ for terms (both variables and constants) of type t , and O, M, N for terms of type $t \rightarrow t$. For constants and variables of predicate types, we often use $F^{\sigma \rightarrow \tau}, G^{\sigma \rightarrow \tau}, H^{\sigma \rightarrow \tau}$ and $X^{\sigma \rightarrow \tau}, Y^{\sigma \rightarrow \tau}, Z^{\sigma \rightarrow \tau}$ respectively, with the explicit typing super-script for disambiguation.

The complex terms of $\mathcal{L}$ are now given by one of two processes: application or abstraction.

- If A has type $\sigma \rightarrow \tau$ and B has type σ , then the application (AB) has type τ and is a complex term of $\mathcal{L}$.

- If A has type τ and v is a variable of type σ , then the abstraction $(\lambda v.A)$ has type $\sigma \rightarrow \tau$ and is a complex term of $\mathcal{L}$.
- Nothing else is a complex term of $\mathcal{L}$.

In contrast to predicate types, we adopt the convention that applications are left-associative, so ABC means $((AB)C)$. We also will write successive λ -abstractions using only one λ and may drop the parentheses, so by $\lambda xyz.A$ we mean $(\lambda x.(\lambda y.(\lambda z.A)))$.

λ -abstraction is one of the largest sources of difference between higher-order logic and first-order logic, so it deserves some explanation. Historically, the λ -calculus was introduced by Alonzo Church in the 1930's as a model of computation based off of substitution. λ -abstractions in this context represent functions where the λ operator binds a variable in the body of the expression that the λ is abstracting over. Application is then the process of applying that function to an argument. This is done through what is called “ β -reduction”, which is simply the process of uniformly substituting the argument into the body of the expression for each occurrence of the bound variable. For instance, the application $(\lambda x.xax)b$ will β -reduce to the expression bab by uniformly substituting b for each bound occurrence of x . This is the heart of the λ -calculus as a model of computation, but our use of λ 's in $\mathcal{L}$ is slightly different.

In $\mathcal{L}$, λ -abstraction is the process by which we build complex predicates and application is how we grammatically apply those predicates. For instance, $\lambda x.Fx$ is a complex term of type $e \rightarrow t$ which we can think of at first blush as expressing something like “the property of being F ” (much more will be said about this when we discuss the semantics of our logic). If a is a term of type e , then the application $(\lambda x.Fx)a$ expresses the predication of ‘being F ’ to a . Note that this expression is not itself β -reduced as in the example from the above paragraph. A form of β -reduction will come out later when we present our proof system.

Officially, λ -abstraction is the only mechanism in our language which binds variables, and we adopt the standard definitions of scope, bound vs. free occurrences, and equating expressions which differ only up to choice of bound variables which are familiar from working

with quantifiers in first-order logic. One strange consequence of this choice is that our quantifiers $\forall_\sigma$ and $\exists_\sigma$ don't actually bind variables themselves. Instead, they attach to a λ -abstraction and ‘take over’ the variable bound by the λ . For instance, instead of writing $\forall_e x(x = x)$, we should write $\forall_e(\lambda x.x = x)$. As noted by (Dorr et al., 2021, p. 21), this way of expressing quantification is a regimentation of Frege’s insight that quantifiers act as higher-order predicates which take other predicates as arguments. Indeed, the reason $\forall_e(\lambda x.x = x)$ is a well-formed complex term of type t is because $\forall_e$ is of type $(e \rightarrow t) \rightarrow t$, $(\lambda x.x = x)$ is of type $e \rightarrow t$, and hence by application $\forall_e(\lambda x.x = x)$ has type t . Yet still, many may find this treatment of quantifiers unintuitive and unfamiliar. So we adopt the convention of writing our quantifiers as explicitly binding variables, as in $\forall_e x(x = x)$.

A similar, and final, convention relates to the logical constants. Officially by the rule of application, we should write $p \wedge q$ as $((\wedge p)q)$. This notation system has historically been referred to as “Polish notation” due to its use by the Polish school of logicians in the early and mid 20th century. But to most people Polish notation is difficult to work with and confusing to look at. So, we adopt the convention of writing the binary logical constants “infix”, exactly how we write them in first-order logic.

This concludes our description of the syntax of our higher-order language $\mathcal{L}$. In summary: after fixing a type system to match the grammar of the language, we introduced various simple terms composed of constants and variables at each type, and showed how to use them to build up complex terms using λ -abstraction and application. Next, we present a proof system for $\mathcal{L}$ which fixes the higher-order logic $\mathcal{H}$.

2.2 The Higher-Order Logic $\mathcal{H}$

Formally, the higher-order logic $\mathcal{H}$ is a particular subset of the formulas (terms of type t) of $\mathcal{L}$ which is closed under a set of inference rules. As was mentioned in the introduction to this chapter, the formulas of $\mathcal{H}$ are intended to be the logical truths of $\mathcal{L}$. We will present the logical axioms and inference rules of $\mathcal{H}$ as a “Hilbert-style” axiomatic proof system. In

axiomatic proof systems, the logical axioms are *presented* as schemes using meta-language variables to stand for arbitrary terms of a fixed type. But the actual axioms contained in the logic are the valid substitution instances of terms for these meta-language variables. $\mathcal{H}$ is then defined to be the smallest set of formulas containing those substitution instances and closed under the inference rules. We write $\vdash \phi$ to say that ϕ is a formula of $\mathcal{H}$.

We will first present the axiom schemes of $\mathcal{H}$. The axiom schemes may be divided into two kinds: generalizations of those from first-order logic and those unique to λ -terms. In these axiom schemes, ϕ is an arbitrary formula of the language $\mathcal{L}$; ψ is a term of type τ ; ξ has type $\sigma \rightarrow t$; δ has type $\sigma \rightarrow \tau$; and α , β , and ν have type σ . The use of Greek symbols is to remind us that they are meta-language variables. Finally, $\psi[\alpha/\nu]$ is the result of uniformly substituting α for each bound occurrence of ν in the expression ψ in the same manner as in our discussion of β -reduction above.

The axiom schemes of $\mathcal{H}$ are the following.

PC $\vdash \phi$ when ϕ is a tautology from classical propositional logic

UI $\vdash \forall_\sigma \xi \supset \xi \alpha$

EG $\vdash \xi \alpha \supset \exists_\sigma \xi$

Ref $\vdash \alpha =_\sigma \alpha$

LL $\vdash \alpha =_\sigma \beta \supset (\xi \alpha \supset \xi \beta)$

β -Eq $\vdash \phi[(\lambda \nu. \psi) \alpha] \leftrightarrow \phi[\psi[\alpha/\nu]]$

The first five axiom schemes are the generalizations of those from first-order logic. **PC** tells us that $\mathcal{H}$ contains each classical tautology, such as $Fa \vee \neg Fa$, while **UI** and **EG** are the corresponding higher-order versions of universal instantiation and existential generalization. **Ref** and **LL** ensure that the equality symbol $=_\sigma$ at each type σ behaves in the way one would expect viz. reflexivity and Leibniz's Law. **β -Eq** (“ β -equivalence”) is the axiom scheme which tells us how λ 's behave and it deserves some explanation.

Consider the formula $(\lambda x.Fx)a \leftrightarrow Fa$ which is a valid substitution instance of $\beta\text{-Eq}$. If $\beta\text{-Eq}$ only contained the left-to-right direction, then it would yield the principle of β -reduction we discussed in the previous section: an application implies its reduced form. But $\beta\text{-Eq}$ also contains the right-to-left implication, which tells us that we may go backwards from a “reduced” expression to an application. This converse direction is often called “ β -expansion”. What is important to note is that β -expansion is non-deterministic, in the sense that the same reduced expressions may β -expand into multiple applications. For instance, $(\lambda X.Xa)F \leftrightarrow Fa$ is also a valid substitution instance of $\beta\text{-Eq}$. Combining the above two instances of $\beta\text{-Eq}$ with **PC** we get $\vdash (\lambda X.Xa)F \leftrightarrow (\lambda x.Fx)a$ which would not be true if we only had the left-to-right direction given by β -reduction. This example shows that $\beta\text{-Eq}$ is a very strong axiom scheme compared to β -reduction alone.

It is also important to note that $\beta\text{-Eq}$ is more general than the above example might imply. Consider the formula $((\lambda x.Fx)a \wedge Ga) \leftrightarrow Fa \wedge Ga$ which is also a valid substitution instance of $\beta\text{-Eq}$. The generality of $\beta\text{-Eq}$ is that it allows β -reduction and expansion *within* some surrounding formula, not just for λ -expressions on their own. To see this, compare $\beta\text{-Eq}$ to a weaker axiom scheme (“extensional β ”)

$$\mathbf{E}\beta \vdash (\lambda \nu.\psi)\alpha \leftrightarrow \psi[\alpha/\nu]$$

which does not allow β -reduction and expansion within the context of a wider formula. While $(\lambda x.Fx)a \leftrightarrow Fa$ and $(\lambda X.Xa)F \leftrightarrow Fa$ are both instances of $\mathbf{E}\beta$, $((\lambda x.Fx)a \wedge Ga) \leftrightarrow Fa \wedge Ga$ is not. The strength and generality of $\beta\text{-Eq}$ will feature heavily in the arguments to be presented in later chapters so they are worth keeping in mind.

Next, we will give the inference rules for $\mathcal{H}$. Each inference rule is a straightforward generalization of those from modal and first-order logic. In these inference rules, ϕ and ψ are both formulas while ν is a variable of type σ .

MP If $\vdash \phi$ and $\vdash \phi \supset \psi$, then $\vdash \psi$.

Gen If $\vdash \phi \supset \psi$ and ν does not occur free in ϕ , then $\vdash \phi \supset \forall_\sigma \nu \psi$

Ins If $\vdash \phi \supset \psi$ and ν does not occur free in ψ , then $\vdash \exists_\sigma \nu \phi \supset \psi$

Nec If $\vdash \phi$ then $\vdash \Box \phi$.

We may now define the logic $\mathcal{H}$ to be the smallest set of formulas of $\mathcal{L}$ containing each of the logical axioms and closed under the inference rules. A proof that $\vdash \phi$ for some formula ϕ is a finite sequence of formulas, the last of which is ϕ , such that each formula in the sequence is either a substitution instance of one of the axioms or else follows from the above formulas in the sequence via one of the inference rules. The connection between these two ways of phrasing things means that asking if $\vdash \phi$ is equivalent to asking whether there exists some proof of ϕ using only the above axioms and inference rules.

Before moving on to the semantics of $\mathcal{H}$, it is worth explaining one important consequence of how $\mathcal{H}$ was defined. Combining β -**Eq** and **EG**, we are able to prove each instance of what is called a “higher-order comprehension scheme”. The term “comprehension” comes from an analogous axiom in set theory. The axiom of comprehension in set theory tells us that for any definable predicate ϕ there exists a set S such that $x \in S$ iff $\phi(x)$. Of course, this version of comprehension is inconsistent, as was shown by Russell in his famous letter to Frege. The contemporary set-theoretic solution to this inconsistency is to restrict which definable predicates ϕ we are allowed to use and to always require some initial set from which we pull the elements x , giving us the “restricted comprehension axiom”.

Our higher-order comprehension scheme is a similar principle which tells us that for any formula ϕ in which X does not occur free, there exists some X of type $\sigma \rightarrow t$ such that $X\nu$ iff $\phi(\nu)$ is true. That is, $\mathcal{H}$ proves every instance of the following scheme.

(COMPREHENSION) $\exists_{\sigma \rightarrow t} X \forall_\sigma v (Xv \leftrightarrow \phi(v))$

It turns out that if we take COMPREHENSION as an axiom scheme instead of β -**Eq**, we are able to prove each instance of **E β** (the weaker β axiom), and moreover the converse is also true. The fact that COMPREHENSION is derivable in $\mathcal{H}$ from β -**Eq** will be of central importance later, so I will end this section by giving its formal derivation and explaining the general idea behind it. This will be our only formal proof we go through in full detail.

For ease of presentation, we will drop the typing annotations and use the variable conventions mentioned above. Now, let $\phi(v)$ be an arbitrary formula with a free occurrence of the variable v . The following is a formal proof in $\mathcal{H}$ of our COMPREHENSION scheme.

1. $(\lambda x.\phi(x))v \leftrightarrow \phi(v)$ (β -Eq)
2. $p \vee \neg p$ (PC)
3. $(p \vee \neg p) \supset ((\lambda x.\phi(x))v \leftrightarrow \phi(v))$ (PC)
4. $(p \vee \neg p) \supset \forall v((\lambda x.\phi(x))v \leftrightarrow \phi(v))$ (Gen 2)
5. $\forall v((\lambda x.\phi(x))v \leftrightarrow \phi(v))$ (MP 2, 4)
6. $(\lambda X.\forall v(Xv \leftrightarrow \phi(v)))((\lambda x.\phi(x))v) \leftrightarrow \forall v((\lambda x.\phi(x))v \leftrightarrow \phi(v))$ (β -Eq)
7. $(\lambda X.\forall v(Xv \leftrightarrow \phi(v)))((\lambda x.\phi(x))v)$ (MP 5, 6)
8. $(\lambda X.\forall v(Xv \leftrightarrow \phi(v)))((\lambda x.\phi(x))v) \supset \exists X(\forall v(Xv \leftrightarrow \phi(v)))$ (EG)
9. $\exists X(\forall v(Xv \leftrightarrow \phi(v)))$ (MP 7, 8)

Informally, the derivation may be separated into two main ideas, one happening in 1-5 and the other in 6-9.

The goal of steps 1-5 is to prove the universally quantified statement $\forall v((\lambda x.\phi(x))v \leftrightarrow \phi(v))$. We do this by starting with an instance of β -equivalence and then universally generalizing over it. Steps 2 and 3 may look extrenuous, but they are necessary to set up the general form that **Gen** wants. 5 concludes the desired formula with a use of modus ponens.

The goal of steps 6-9 is similar in that it wants to prove the existentially quantified statement $\exists X(\forall v(Xv \leftrightarrow \phi(v)))$, which is exactly our instance of COMPREHENSION. We again use an instance of β -equivalence to set up our formula in the way that **EG** wants, and then we use modus ponens to conclude our instance of COMPREHENSION.

This concludes our discussion of the proof system for our higher-order logic $\mathcal{H}$. To summarize: we set up an axiomatic proof system which singles out the sentences of $\mathcal{L}$ which we take to be the logical truths. Such a sentence is in $\mathcal{H}$ just when we can exhibit a finite proof of that sentence using the axioms and inference rules. In the next section, we will work towards an understanding of what the intended *meanings* of these sentences are.

2.3 The Intended Interpretation of $\mathcal{L}$

So far we have presented the higher-order logic $\mathcal{L}$ and shown how a particular proof system singles out its logical truths $\mathcal{H}$. But in doing so we have ignored a central question: what is the semantic interpretation of our language? That is, what do its formulas mean? It is difficult to see the justification for calling the formulas in $\mathcal{H}$ logical truths if we haven't discussed what an interpretation of the symbols of our language should be.

In fact, higher-order languages like $\mathcal{L}$ can be given a set-theoretic model theory, just like first-order languages, which is full of rich and interesting mathematics. Once this is done, soundness and completeness results may be proved for particular classes of models which justify the claim that the formulas of $\mathcal{H}$ are exactly those formulas true regardless of how the terms of the language are interpreted. But we will not be concerned with model-theoretic interpretations.

As Williamson (2013) argues, there is an *intended* semantics of our language.³ The constant symbols $\neg$, $\wedge$, $\vee$, and $\supset$ are intended to mean logical negation, conjunction, disjunction, and material implication, respectively. Moreover, we employ the principle of compositionality: the intended meaning of a complex term is built up from those of its constituents and how they are combined. $=^{e \rightarrow e \rightarrow t}$ is intended to mean the identity relation among objects, that relation which holds between a and b just when a and b are one and the same. The quantifiers $\forall_e$ and $\exists_e$ are intended to mean objectual universal and existential quantification over the domain of objects. $\Box$ is intended to mean metaphysical necessity. That is, a formula in our language of the form $\Box\phi$ is intended to mean that, with respect to the modal profile of our reality, ϕ could not possibly have been false. So far this matches the intended interpretation of a first-order modal logic. But what, then, are the intended meanings of our higher-order constructions such as predicates, higher-order quantifiers, and λ -abstractions?

One interpretation is that higher-order quantification is a restricted form of first-order

³It should be noted that Williamson's goals are not to investigate the intended interpretation, but instead to investigate a halfway point between that interpretation and validity in a fully general model theory.

quantification and the referents of higher-order terms are abstracta included in our first-order domain. On this view, we have in our intended interpretation certain abstract objects, namely properties, which act as the referents of predicates and λ -abstractions. A quantifier such as $\forall_{e \rightarrow t}$ is then understood as a restricted quantifier over the subset of our domain containing just properties. This is more-or-less what Quine (1986) argues one should do if one is foolish enough to countenance talk of properties in the first place, something higher-order logic is designed to do. Yet proponents of higher-order metaphysics argue that not only is this reduction of higher-order quantification unnecessary, but, worse, it is wholly misleading. The key principle of higher-order metaphysics is that higher-order quantification is a *sui generis* form of quantification, one which does not reduce to a restricted form of first-order quantification.

The history of this interpretation of higher-order quantification begins with Frege's distinction between concepts and objects and is most explicitly articulated in Frege (1892a).⁴ Concepts are what Frege calls the 'referents' (*Bedeutungen*) of terms of type $e \rightarrow t$, while objects are the referents of terms of type e . I use scare-quotes in the previous sentence because, as Frege argues forcefully, the every-day way of thinking about reference involves singular terms, terms of type e , and so use of that word to describe what higher-order quantifiers quantify over invites confusion between concepts and objects. Frege is rejecting the idea outlined above, that concepts should be identified with abstract properties in a first-order domain. Concepts are the 'referents' of predicates only insofar as they play a structurally similar role in the semantic meaning of predicates as objects do in the semantic meaning of singular terms. But we should not conflate this way of expressing the situation with the idea that concepts are first-order, abstract properties (or that they are the set of all things which fall under the concept, for that matter). Frege's thesis is that concepts and objects are of fundamentally distinct ontological statuses, a distinction which goes far beyond that

⁴Here I am using 'concept' and 'property' as synonyms for the sake of ease of presentation. Although 'concept' might have different philosophical connotations than 'property' in contemporary metaphysics, Frege's use of 'concept' (*Begriff*) maps more accurately onto the contemporary use of 'property'.

between concrete and abstract. On this view, it is a category mistake to try to compare an object and a concept, and hence no concept is an object or vice versa.

The difficulty in holding this position, as Frege is well aware, is that even expressing the distinction between concepts and objects seems to already involve us in a contradiction. This is manifested in what is now known as “The concept *horse* paradox”.⁵ The problem is that “the concept *horse*” seems to be the English equivalent of the higher-order term $\lambda x.\text{Horse}(x)$, a predicate term of type $e \rightarrow t$, and so should refer to a concept. Yet grammatically “the concept *horse*” is a determiner-phrase, so it is of type e . It follows from Frege’s thesis, then, that the referent of “the concept *horse*” is an object, not a concept. Yet how can this be if the concept *horse* is plainly a concept? The paradox is that, try as we might to express the concept-object distinction, we are compelled by the strictures of natural language grammar to refer to concepts using determiner-phrases and in so doing undermine the distinction we are trying to build. But then, if the conflation of concepts with objects seems unavoidable when attempting to phrase the distinction in natural language, how can we be sure that there is a distinction to begin with?

In response to the paradox, Frege says the following:

I admit that there is a quite peculiar obstacle in the way of an understanding with my reader. By a kind of necessity of language, my expressions, when taken literally, sometimes miss my thought; I mention an object, when what I intended is a concept. I fully realize that in such cases I was relying upon a reader who would be ready to meet me halfway—who does not begrudge a pinch of salt. —
(Frege, 1892a, p. 192)

That is, Frege’s response to the paradox is for us to move past it, to accept that there is a deeper, metaphysical distinction at work here which natural language does not track. Thus, he wants to take a therapeutic, as opposed to direct, route to understanding the object-

⁵“*horse*” is written in italics here to emphasize that we want to express the concept. For a book length treatment of the paradox, see Trueman (2020).

concept distinction. But even supposing that we accept this concession as a solution to the paradox and agree to follow Frege down this route, much more still needs to be done to illuminate the nature of higher-order quantification.

As a strategy for following through with Frege’s therapeutic approach to elucidating the concept-object distinction, 20th century work on higher-order quantification focused on finding non-nominal expressions of natural language into which we may regiment our higher-order vocabulary.⁶ Consider the higher-order statement $\exists X(Xa)$. If we translate this into English as “there exists an X such that a is X ” we fall victim to the concept *horse* paradox since we are regimenting the higher-order variable X into noun-phrase position (“*an X*”). This straightforward translation *is* justified if we take the restricted first-order quantifier reading of higher-order quantification, since in that case the quantifier $\exists X$ is really quantifying over abstract properties (or their extensions) and so we are correct to speak of X as an object. But since the proponent of *sui generis* higher-order quantification is rejecting this interpretation, they may hope for a better strategy in finding some other, non-nominal grammatical construction of natural language into which we can translate higher-order terms such as $\exists X$.

Rayo and Yablo (2001), building on the earlier work of Prior (1971) and Boolos (1984), argue that this “denominalization” of higher-order terms may be achieved by instead regimenting higher-order variables into *adverb* position. On this strategy, a better translation of $\exists X(Xa)$ is “ a is somehow”. They argue that this adverbial translation strategy better captures the *sui generis* interpretation of higher-order quantification in two important respects. The first is that this translation strategy reflects that higher-order variables are meant to be *predicative*. Since “somehow” is adverbially modifying the copular “is” to produce a predicative verb-phrase, as opposed to the nominal phrase “an X ”, this translation strategy yields

⁶By “non-nominal” I mean moving away from grammatically nominal expressions, like noun-phrases. This discussion may be confusing since “non-nominal” could be interpreted as opposition towards philosophical nominalism, as in the realism vs. anti-realism sense. There *is* a relationship between a nominalist (anti-realist) motivation for *sui generis* higher-order quantification and the (grammatical) non-nominalist translation strategy, as the title of Rayo and Yablo (2001) alludes to (“[Metaphysical] Nominalism Through [Grammatical] Denominalization”). But it is important to keep the two uses of ‘nominal’ separated here to avoid confusion.

a grammatically correct mapping between the $e \rightarrow t$ predicate type term X and the predicative verb phrase “is somehow”. The second point is that this form of translation correctly captures key defining aspects of quantification such as scope and anaphoric cross-indexing. For example, on this strategy we should translate $\exists X(Xa \wedge Xb)$ as “ a is somehow, and so is b .” In this sentence, the anaphoric dependence of “so” on “somehow” correctly captures that $\exists X$ has scope over both occurrences of X . Both of these points combine to provide hope that this non-nominal translation strategy can help us to get a grip on *sui generis* higher-order quantification while avoiding the concept *horse* paradox.

21st century work in higher-order metaphysics, however, has moved away from the above strategy of translation. One reason for this is that the translation strategy was deployed only to handle quantification over second-order predicates (those of type $e \rightarrow t$) and it is not clear how this should be generalized along the infinite hierarchy of predicate types. That is, English doesn’t seem to give us immediate access to a familiar hierarchy of ad-adverbs, ad-ad-adverbs, etc., which allow for a structural mapping from our higher-order terms. And even if we did find success in first translating third-order and then fourth-order and then ..., it is difficult to shake the feeling that there is something fundamentally wrong with such a piecemeal strategy.

Contemporary proponents of higher-order metaphysics such as Williamson (2003) and Dorr (2024) argue that even if there is no one-to-one mapping between our higher-order language and a natural language such as English, this should not count as strong evidence against the possibility of understanding a *sui generis* interpretation of higher-order quantification. This is because there are many pieces of first-order, and even propositional, logic which similarly resist direct mapping into natural language. As Williamson and Dorr argue, the material conditional is one such example. Although we do gain intuition for reasoning with $\supset$ by analogies with the English “if ..., then...”, it is well known that these analogies eventually fail. Yet despite this lack of perfect synonymy, one day the in-training logician does wake up with a full grasp of how $\supset$ works. Williamson likens this to the immersion

strategy, or direct method, we often use to learn a new natural language. And this direct method is not as mysterious as it seems, as Dorr points out, since in the case of logic a straightforward strategy for gaining such an understanding comes in the form of learning about the *inferential role* $\supset$ plays in the language of propositional logic (Dorr, 2024, pp. 9–10). We learn how $\supset$ works by learning how to linguistically reason with expressions involving $\supset$, not by directly mapping it to “if ..., then ...”. And if this is a strategy available for other logical languages, why should we assume it is not possible for $\mathcal{L}$?

For the sake of engaging with the higher-order metaphysics project on its own terms, I will take on board this understanding of higher-order quantification. I do find the view to be *plausible*, even if it is clear to me why many do not. But for those who find the view *incoherent*, I will offer a small argument to push them in the opposite direction.

Sui generis interpretations of quantification are not unique to higher-order languages. In fact, they form a core part of the philosophy of logic. The debate between objectual quantification and substitutional quantification as interpretations of the first-order quantifiers is a debate of the same family, and it played a crucial role in the early development of first-order modal logic through the work of Ruth Barcan-Marcus. Similarly, if debates such as those between classical and intuitionistic readings of the logical connectives, or between pluralism and monism about logical consequence, are at all coherent, it is because we do have a grasp on what it means to give an alternative interpretation of logical vocabulary. And most do take those debates to be not only coherent, but philosophically fruitful. So, we should separate skepticism of giving *any* coherent *sui generis* interpretation of higher-order logic from skepticism that the particular interpretation outlined above is coherent. While the former skepticism is unwarranted, I sympathize with the latter in my desire for higher-order metaphysicians to say much more about the nature of the interpretation than they do. Yet still, I do not take their silence as a self-refutation. If their position is correct, then it is easy to see why something like the direct method is needed for gaining familiarity with the intended interpretation. This, I hope, is enough to find the view at the very least coherent.

Supposing that the direct method for grasping a *sui generis* interpretation of higher-order quantification is a coherent strategy, we are still left with the problem of supplying Frege's promised pinch of salt. Philosophy cannot, and should not, be done entirely in artificial languages such as $\mathcal{L}$. So we will need to provide *glosses* for higher-order terms into English to facilitate communication. I will follow the standard set by Williamson, Dorr, and others. I will use 'entity' to refer to the *Bedeuten* of terms of our language of any type, 'property' for those of terms of type $\sigma \rightarrow \tau$, 'objects' for those of terms of type e , and 'propositions' for those of terms of type t . I will also gloss formulas such as $\exists X(Xa)$ in the naive, nominal way critiqued above. But, as I have stressed in this section, this is merely for ease of communication. These English glosses of our higher-order terms should be understood as an informal mode of speaking; when we wish to be precise, we use terms of $\mathcal{L}$. When so understood, it should be clear that this way of speaking is not committing ourselves to the existence of first-order abstracta, since it is an obvious mistake to read ontological commitments off of informal ways of speaking.

Finally, then, we have an answer to our initial question about the intended interpretation of our higher-order language. The terms of $\mathcal{L}$ describe a higher-order metaphysical reality. The types of our language guide ontological distinctions between the objects that exist in the world and the ways in which they exist. Accepting the central thesis of higher-order metaphysics is accepting the claim that this higher-order metaphysical reality is in fact the reality we live in. Moreover, the formulas of $\mathcal{H}$ are exactly the logical truths of this reality. But this means $\mathcal{H}$ will only describe *generalizations* of our metaphysical reality, things which hold no matter how we interpret our non-logical constants and symbols. So on its own, $\mathcal{H}$ is not enough to completely fix all truths of metaphysics expressible in $\mathcal{L}$. For that, we need to build a theory on top of $\mathcal{H}$, one which will capture all of the truths of our intended interpretation. Beginning in the next chapter, we will move towards this goal by considering further axioms which fix the behavior of higher-order identity statements between propositions.

3 Higher-Order Theories of Grain

The search for the identity criteria of propositions has been a central problem in analytic philosophy since its beginning. Frege (1892b) introduced the famous example sentences “Hesperus is Hesperus” and “Hesperus is Phosphorus” to develop a theory of how informative identity statements, such as the second, could be possible. Frege’s diagnosis was, roughly, that the two distinct names ‘Hesperus’ and ‘Phosphorus’ contribute different semantic values to the respective sentences, despite referring to the same object, and as a result the propositions expressed by the two sentences are distinct as well. And if the propositions are distinct, we shouldn’t be surprised that one may contain non-trivial information even if the other does not. What falls out of Frege’s theory, then, is an account of the identity conditions of propositions which tells us that mere co-designation of constitutive singular terms is not sufficient for propositional identity. Many other theories of propositional identity have been proposed since and the question remains a central topic in both metaphysics and the philosophy of language.

Dorr (2016)’s project is to bring the tools and methodology of higher-order metaphysics to bear on these so-called “identification” questions, questions concerning the identity of higher-order entities such as propositions. Importantly, given the intended interpretation of our higher-order language, these questions are similar in form (in that they ask about the behavior of the $=_\sigma$ symbols) but still distinct questions from identity questions between objects. Dorr considers not just propositions, but also identification questions for properties at each higher-order predicate type $\sigma \rightarrow \tau$. It turns out that we may remain neutral to the other higher-order identity questions while still giving complete answers to the identification question of propositions, and so for the rest of this thesis we will consider only propositions. Roughly, then, our project is to ask what are the necessary and sufficient conditions under which we may infer that the proposition p is identical to the proposition q .

An immediate benefit of framing this question in the context of higher-order metaphysics is that we can rigorously express our goal as one of building a theory on top of our logic $\mathcal{H}$.

Recall that for a sentence ϕ in a formal language, we say that ϕ is *undecided* by a theory (or a logic) just when neither ϕ nor $\neg\phi$ is a sentence of the theory. With respect to our higher-order logic $\mathcal{H}$, we would then say that ϕ is undecided by $\mathcal{H}$ just when neither $\vdash \phi$ nor $\vdash \neg\phi$, that is, there does not exist a proof of ϕ from the axioms of $\mathcal{H}$ and there does not exist a proof of $\neg\phi$ from the axioms of $\mathcal{H}$. Since our intended interpretation of our language $\mathcal{L}$ fixes the truth of every sentence ϕ , one and only one of ϕ or $\neg\phi$ is true. But then, if ϕ is undecided by our logic $\mathcal{H}$, we need to add additional axioms to capture that truth.

It turns out that $\mathcal{H}$ is surprisingly neutral when it comes to such statements and indeed there are many propositional identifications which $\mathcal{H}$ leaves undecided. As a concrete example, consider the sentence $\forall_t p.p =_t (p \wedge p)$ which we may gloss as “every proposition p is identical to $p \wedge p$ ”. There are prominent theories of propositions which do hold that every proposition is identical to its self-conjunction. Such a theory, when expressed in our higher-order language, would count $\forall_t p.p =_t (p \wedge p)$ as one of its theorems in addition to each of its universal instantiations. Yet as it turns out both the universal statement and each of its instantiations is undecided by $\mathcal{H}$.⁷ The upshot, then, is that providing a higher-order theory of propositional identity amounts to proposing axioms governing the use of $=_t$ which would decide every statement involving it. Arguing for our theory similarly amounts to arguing for the adoption of those axioms.

In the next section I will give a preliminary argument for a particular theory of propositions, a so-called *structured* theory which has its roots in Frege and Russell, and show what such a theory looks like in our higher-order language $\mathcal{L}$. In the following section I will present the Russell-Myhill paradox, which is presented in much of the higher-order metaphysics literature as presenting a decisive argument against the structured theory. The goal of Chapters 4 and 5 is to defend against this claim.

⁷See Bacon (2024) or Dorr (2016) for formal proofs of the metalogical results claimed in this chapter.

3.1 An Argument for Fine-Grained Propositions

One way that the identification questions of higher-order entities has been phrased is: “how fine-grained is reality?” Theories which posit many identifications between higher-order entities are often said to be *coarse-grained* theories, since they claim that the metaphysical divisions of reality are few and far between. On the other hand we have *fine-grained* theories which posit many distinctions between higher-order entities. Given their affinity for this way of phrasing the problem, some higher-order metaphysicians, such as Goodman (2024), refer to this project as offering higher-order theories of grain, or as “grain science.”

In this section, I want to offer an argument for a particularly fine-grained theory of propositions, one which is often referred to as the *structured theory of propositions*. The core idea behind the structured view is that propositions are in some sense complexes which contain constituents and as a result the identity criteria for propositions are sensitive to the identities of those constituents. Before presenting the higher-order version of the structured view I will go on to argue for, I will first outline its history.

Frege’s theory sketched above, as well as the theory offered by Russell (1903), are considered early versions of a structured theory of propositions. Frege’s theory relies on his distinction between sense and reference. On this view, there are only two referents of sentences: the True and the False. But each sentence also has a sense which determines *how* it refers to its referent, and the sense of a sentence is the proposition it expresses. Propositions on this view are complexes which are made up of the senses of their constituent expressions and the ways in which these constituent senses are put together. This is why co-designation of terms is not sufficient for the identity of propositions as noted above: the terms may have the same referent yet different senses. Russell held a more direct view about propositions: propositions are abstract complexes which contain the very objects the propositions are about.

Many theories of structured propositions presented over the last century are refinements in one way or another on either the Frege or Russell view. With one notable exception in

Church (1951), most of those in the Fregean camp find senses to be mysterious and try to replace talk of senses with talk of intensions—functions from possible worlds to truth values, properties, or objects. This approach to Fregean theories of structure began with Carnap (1947)’s notion of intensional isomorphism and found mature expressions in Lewis (1970) and Cresswell (1985). The first worked-out reconstruction of the Russellian view is due to Kaplan (1975) and remains the more popular view today. King et al. (2014) is a dialectic between three of the most eminent Russellian theorists each arguing for their own preferred view of the truth-bearing and representational nature usually attributed to propositions.

With Frege and Church being the two exceptions, what all of the theorists of structured propositions I’ve mentioned above have in common is that they formulated their theories in first-order logic. To the higher-order metaphysician, this is a problem. As I mentioned in Chapter 2, the higher-order metaphysician believes this is at best a misleading and at worst a false way of describing the situation. Fortunately, our higher-order language $\mathcal{L}$ makes My strategy to argue for a higher-order theory of structure is to show what goes wrong with the more coarse-grained alternatives, that is, why we would want a fine-grained theory in the first place, and then to demonstrate why $\mathcal{L}$ provides a particularly elegant way of expressing the structured view. So, to begin, I will first present a view at the opposite end of the coarse-fine grained spectrum.

A very coarse-grained theory of propositions is one which makes many identifications between propositions. Perhaps the most coarse-grained view possible, while still respecting a distinction between true and false propositions, is one which identifies all true propositions as one and all false propositions as another. This view is often called *extensionalism*, since the only distinction it makes between propositions is their extensional truth-values. An extensionalist theory of propositions would then add the following axiom to $\mathcal{H}$.⁸

(EXTENSIONALISM) $\forall p \forall q ((p \leftrightarrow q) \rightarrow p = q)$

Informally, EXTENSIONALISM holds that identifications of propositions are made by material

⁸Starting now, I will often suppress typing subscripts for readability.

equivalence: if p and q are both true (or both false) then for p to hold is for q to hold.

Now, EXTENSIONALISM strikes many philosophers as obviously false. But the principle has had its defenders. For instance, EXTENSIONALISM is the most plausible way of reading proposition 5.141 from the *Tractatus*: “If p follows from q and q from p , then they are one and the same proposition,” (Wittgenstein, 1922, p. 58). EXTENSIONALISM also has a complicated history of being associated with Frege, since he argued that there are only two *referents* of sentences, the Truth and the False, and in fact EXTENSIONALISM is occasionally referred to as the “Fregean Axiom” (see, e.g., Bacon (2024)). The fine-grained structured view of Frege is then recovered via complicating the logic with senses in addition to referents, but let it pass. In light of EXTENSIONALISM’s history, it is worth offering an argument for why it is an implausible theory of propositional identity.

First, it is important to remember that propositions as theoretical entities have been associated with the meanings of sentences since, at the very least, Frege. As such, propositional identifications are meant to answer questions such as the following. Suppose that Audrey uttered the sentence s at time t . Suppose further that it is unquestionably true that s expresses the proposition p . Did Audrey’s utterance of s express q ? Supposing that p is a true proposition, EXTENSIONALISM predicts that Audrey’s utterance did express q for any other q which happens also to be true. But this result cannot be correct, because it fails to recognize that sameness of truth is insufficient for sameness of meaning. If propositions truly serve as the meanings of sentences, then EXTENSIONALISM predicts that all communication is a trivial back-and-forth between the True and the False. Yet even casual communication is a rich exchange of nuanced, intricate information. So EXTENSIONALISM cannot be true on pain of being *too* coarse-grained: it predicts too many propositional identifications for the theoretical use in explaining communication that propositions are widely held to have.

Next, we will consider a view which is often thought to sit in the middle between coarse-grained and fine-grained views. The view may be motivated by noticing that the extensionalist view gets something right when it ties propositional identity to truth, it just does

so too coarsely. The problem with EXTENSIONALISM on this view is that it isn't sensitive to *all possible* truth conditions of p and q and instead focuses on which truth values they happen to have. If p and q have the same truth value in all possible evaluations, the view goes, then we are correct to identify them. This view is often called *intensionalism* since it is the intensions of the propositions, viewed as functions from possible worlds to truth values, which are used to identify them. Since our language $\mathcal{L}$ contains the modal operator $\Box$, we may formalize the intensionalist view as adding the following axiom to $\mathcal{H}$.

(INTENSIONALISM) $\forall p \forall q (\Box(p \leftrightarrow q) \rightarrow p = q)$

Informally, INTENSIONALISM says that identifications of propositions are made by *necessary* equivalence: if it is necessary that p and q are both true (or both false), then for p to hold is for q to hold. Importantly, the converse of INTENSIONALISM holds as a general theorem of $\mathcal{H}$. This means that $\mathcal{H} + \text{INTENSIONALISM}$ gives both necessary and sufficient conditions for the use of $=_t$ and so is a strong enough theory to decide every propositional identity statement in our language $\mathcal{L}$.

Intensionalist views of propositional identifications are among the most popular in the higher-order metaphysics literature, and it represents the main alternative to the structured view I will go on to defend. As such, the main arguments for the view will emerge over the next few chapters. For now, I will offer what I take to be the most direct argument, first directly offered by Soames (1987), against INTENSIONALISM which leads to the structured view.

The best way of seeing what is wrong with INTENSIONALISM is by unpacking what the statement means in terms of the standard possible worlds semantics (SPWS) of modal logic. In SPWS, the antecedent claim $\Box(p \leftrightarrow q)$ is true iff $p \leftrightarrow q$ is true at all possible worlds. In other words, the antecedent is true if and only if $p \leftrightarrow q$ is itself a necessary truth. The problem is that $p \leftrightarrow q$ will be a necessary truth if p and q are both themselves necessary statements: any two necessary statements have the same truth value at every possible world since they are always true in any possible world. Take for example “all birds are birds”

and “ $2 + 2 = 4$ ”. Both of these statements under standard assumptions about vacuous and mathematical truth are necessarily true. But then we must have that “necessarily, all birds are birds if and only if $2 + 2 = 4$ ” is true as well, and so INTENSIONALISM predicts that the two statements express the same proposition. In fact, one of the main theoretical results of INTENSIONALISM is that there is one and only one necessary proposition. This does not seem right.

What INTENSIONALISM fails to capture is that two necessary propositions may be *about* different things. Although what “all birds are birds” expresses is a trivial fact, it is a fact *about* birds, and not about apples, or oranges, or numbers, or the moon. INTENSIONALISM does not leave room for such a distinction to be drawn and as such it is forced to collapse any two propositions with the same modal profile into one. Even worse, this collapse threatens to trivialize the distinction between any two true mathematical statements. Is it really true that “ $2 + 2 = 4$ ” and, say, Fermat’s Last Theorem express the same proposition? Was Andrew Wile’s proof of the Theorem, widely celebrated as one of the most ingenious and groundbreaking pieces of mathematics humans have ever achieved, just another way of proving that $2 + 2 = 4$? Are all mathematical proofs, no matter their difference in complexity and background assumptions, just different ways of establishing the single propositional fact which holds in all possible worlds, the same proposition expressed by ‘all birds are birds’? I cannot see how this could possibly be true. Thus, given that INTENSIONALISM fails to capture the crucial fact that propositions are about particulars, we should reject INTENSIONALISM and move toward a more fine-grained theory.⁹

The idea that propositions are in some way about their constituents is exactly the motivation behind adopting the structured view of propositions. Theories of grain which posit distinctions between necessarily equivalent propositions are often called ‘hyperintensional’ theories, and the structured view is one of the most straightforward ways of capturing this idea in our higher-order language. Consider the following axiom scheme, which has one

⁹It may seem that I am moving on without considering possible responses from defenders of INTENSIONALISM. Those responses and my replies to them make up Chapter 4.

instance for each choice of σ .

$$(\text{STRUCTURE}) \quad \forall_{\sigma \rightarrow t} F \forall_{\sigma \rightarrow t} G \forall_{\sigma} x \forall_{\sigma} y (Fx =_t Gy \rightarrow (F =_{\sigma \rightarrow t} G \wedge x =_{\sigma} y))$$

Informally, STRUCTURE says that the proposition Fx is to be identified with Gy only if its constituents F and G as well as x and y are identified. That this axiom scheme captures the aboutness nature of propositions can be seen by taking its contrapositive: if two propositions are built from non-identified pieces then they are not the same propositions. It should be clear, then, that STRUCTURE does capture hyperintensional distinctions between necessary propositions, since, in our above examples, “all birds are birds” and “ $2 + 2 = 4$ ” contain different constituents.

It may be objected that STRUCTURE seems to already assume some sort of hyperintensional theory, since it relies on the use of $=_{\sigma \rightarrow t}$ and $=_{\sigma}$. In fact, there are extensional, intensional, and hyperintensional versions of property identifications, as explored by Dorr (2016), and it would be in some way question-begging against INTENSIONALISM if we assumed a version of hyperintensional distinctions held for every other class of higher-order entities and used that to derive hyperintensionalism for propositions. Fortunately, as I alluded to at the beginning of this chapter, the specific choice of identity criteria for properties does not actually cause a difference here.¹⁰ We may grant the intensionalist a version of INTENSIONALISM for properties and still hold that the identity conditions for propositions are sensitive to their compositional structure, the choice would not collapse STRUCTURE into INTENSIONALISM.

In the next section we will consider what many believe to be the best objection to STRUCTURE as a higher-order theory of propositional identity: $\mathcal{H} + \text{STRUCTURE}$ is formally inconsistent.

¹⁰Again, see Bacon (2024) or Dorr (2016) for the formal proofs.

3.2 The Russell-Myhill Paradox

The inconsistency of (a version of) STRUCTURE was first noticed by Russell and presented informally in Appendix B of Russell (1903). It was later rediscovered by Myhill (1958) where it was proved that the inconsistency is derivable in Church (1951)’s higher-order logic of sense and reference. Consequently, the result has come to be known as the Russell-Myhill paradox.

The paradox is essentially this. $\mathcal{H}$ on its own is able to prove a version of Cantor’s theorem, specifically that there does not exist an injection from the powerset of a set X to the set X itself. But STRUCTURE implies that higher-order predicates may be injective on their arguments, contradicting Cantor’s theorem. Hence, $\mathcal{H} + \text{STRUCTURE}$ is formally inconsistent. For the rest of this section I will walk through an informal derivation of the Russell-Myhill paradox in $\mathcal{H} + \text{STRUCTURE}$ and then diagnose where dialectically the paradox leaves the proponent of structured propositions.

Now, the informal derivation of the Russell-Myhill paradox I will present follows that given by Linnebo (2024). As Linnebo stresses, the informal nature is just for ease of presentation, and formal derivations of the paradox may be found in Dorr (2016), Goodman (2017), and Bacon (2024) among many others. The structure of the derivation I will present is as follows. We first use COMPREHENSION to prove a version of Cantor’s theorem that there is no injection from a powerset to its underlying set. In our derivation, this is done by showing that there cannot be an entity of type $(t \rightarrow t) \rightarrow t$ which is injective. We then show that this form of Cantor’s theorem directly contradicts STRUCTURE.

First, we assume towards contradiction that there is some operator, call it f , with type $(t \rightarrow t) \rightarrow t$ such that f is injective. That is, assume the following sentence is true:

$$\exists f \forall O \forall M (fO =_t fM \rightarrow O =_{t \rightarrow t} M) \quad (1)$$

Second, in all that follows we will use $\phi(p, q)$ as an abbreviation for the following sentence:

$$\phi(p, q) := \exists O(q =_t fO \wedge Op) \quad (2)$$

All occurrences of $\phi(p, q)$ below should be understood as an abbreviated occurrence of $\exists O(q = fO \wedge Op)$, which is itself an open sentence with two free propositional variables p and q . Informally, what $\phi(p, q)$ says is that there exists some operator O such that q is identified with the proposition given by applying f to O and that O applied to p holds.

Next, we want to prove the following proposition:

$$\forall M \forall p (\phi(p, fM) \leftrightarrow Mp) \quad (3)$$

First, let M be an arbitrary operator of type $t \rightarrow t$ and p an arbitrary proposition. For the left-to-right direction, assume that $\phi(p, fM)$ is true. By (2) this means that there exists some operator O such that $fM = fO$ and that Op is true. But then by (1), we must have that $M = O$, and so by **LL** we have that Mp . For the right-to-left direction, assume Mp . Since $fM = fM$ must be true, so is the conjunction $fM = fM \wedge Mp$. But now using **EG** we may existentially generalize over M twice in this conjunction to get $\exists O(fM = fO \wedge Op)$ which is $\phi(p, fM)$. Since M and p were arbitrary, this establishes (3).

Now, we will use **COMPREHENSION** to diagonalize a ‘Liar-like’ operator R . Consider the open formula $\neg\phi(p, p)$, which informally says that p is not related to itself by ϕ . By **COMPREHENSION** we have

$$\exists R \forall p (Rp \leftrightarrow \neg\phi(p, p)) \quad (4)$$

If we let $r := fR$ name the proposition given by applying f to R , then instantiating R and r into (3) gives us

$$\forall p (\phi(p, r) \leftrightarrow Rp) \quad (5)$$

But now we may combine the biconditionals in (4) and (5) to get

$$\forall p(\phi(p, r) \leftrightarrow \neg\phi(p, p)) \quad (6)$$

and if we instantiate r for p in (6) we have

$$\phi(r, r) \leftrightarrow \neg\phi(r, r) \quad (7)$$

contradiction. Hence, our assumption of (1), that there is such an injective f , was false, and so we have derived

$$\neg\exists f\forall O\forall M(fO = fM \rightarrow O = M) \quad (8)$$

This establishes the proof of Cantor's theorem using COMPREHENSION.

We now reach the final step in the Russell-Myhill paradox. Let p be an arbitrary proposition and let $g := \lambda O.Op$ name the operator of type $(t \rightarrow t) \rightarrow t$ which takes an operator of type $t \rightarrow t$ and predicates it of p . Instantiating g and p into STRUCTURE gives us

$$\forall O\forall M(gO = gM \rightarrow (g = g \wedge O = M)) \quad (9)$$

But then existentially generalizing on g gives

$$\exists f\forall O\forall M(fO = fM \rightarrow (f = f \wedge O = M)) \quad (10)$$

directly contradicting (8). Hence, $\mathcal{H} + \text{STRUCTURE}$ is inconsistent.

So much for the formal details of the paradox. What should we make of the result? Many proponents of INTENSIONALISM believe that the paradox *definitively* shuts the door on structured theories of propositions. For example, Peter Fritz claims that “STRUCTURE can be shown to be false by a purely logical argument” (namely, Russell-Myhill), and moreover that “The Russell-Myhill argument shows that propositions cannot be structured,” (Fritz,

2023, p. 53, 55). Similarly, the central thesis of Goodman (2017) is reflected in its title: “Reality is Not Structured.”

I believe that these conclusions are far too hasty. It is true that we cannot accept both $\mathcal{H} + \text{STRUCTURE}$. But these responses take it for granted that $\mathcal{H}$ itself is beyond reproach. From the derivation of the Russell-Myhill paradox we can isolate β -equivalence as the piece of $\mathcal{H}$ which is incompatible with STRUCTURE . So the real position we have found ourselves in is having to make the choice between abandoning STRUCTURE and embracing something like INTENSIONALISM , or else keeping STRUCTURE and revising $\mathcal{H}$ by abandoning β -equivalence.

My goal for the rest of this thesis is to take the latter path. In the next chapter I will present and defend against arguments that STRUCTURE is at odds with standard principles in contemporary metaphysics. In the following chapter I will offer independent arguments against β -equivalence and defend a revised logic which is compatible with STRUCTURE .

4 A Defense of Higher-Order Structure

In order to understand why intensionalists believe we should abandon STRUCTURE instead of making revisions to $\mathcal{H}$, it is important to have a grasp of the distinction made in contemporary metaphysics between *worldly* and *representational* concepts. This is because a central premise in these arguments is that the intuition for STRUCTURE is rooted in representational, not worldly, intuitions, and we have reason to be wary of such intuitions when engaging in metaphysics.

The distinction between worldly concepts and representational concepts is roughly what the words suggest. Representational concepts are concepts which relate to how we as agents represent the world, either through thought or language. Worldly concepts, or sometimes just non-representational concepts, are those which involve the unmediated features of the world on their own terms. Nolan (2014) points out that the exact place where we draw the line is itself a matter of dispute, but concepts such as believing, knowing, information containing, and semantic meaning are often considered to be representational, while metaphysical concepts such as essence, grounding, and causation are often taken to be worldly.

One plausible criterion for a concept being representational as opposed to worldly is the concept's *epistemic* sensitivity. Representations, whether in thought or in language, are tightly connected to the epistemic attitudes we have about the world. We engage in representation when we think and speak about the world, and those thoughts and sentences are the objects of our epistemic attitudes. The non-representational nature of worldly concepts, on the other hand, is thought to be derived exactly from the non-epistemic nature of the concepts. An object's essence is not tied to a particular way of thinking about the object and it is not derived from a particular way of speaking of the object, even though this was a common way of thinking about essence for a large portion of philosophy's history.

This way of separating worldly and representational concepts has its roots in Saul Kripke's *Naming & Necessity* (*N&N*). One of the most celebrated achievements of *N&N* is Kripke's cleaving of metaphysical and epistemic modality. As I alluded to above, many philosophers

have conflated the necessary, the *a priori*, and the analytic, and Kripke does much work in *N&N* to draw a sharp distinction between these three modalities. Kripke's conception of metaphysical necessity is the worldly, non-representational one outlined above. Whether a proposition is metaphysically necessary is a matter of the worldly claim the proposition makes, irrespectively of any representational aspects of the way we choose to express the proposition. On the other hand, epistemic necessity is representational. Whether a proposition is *a priori* or *a posteriori* may be sensitive to mind-dependent and linguistic context. The proposition, when expressed in some particular language or thought of by some particular agent, will involve representational features which determine whether it may be known *a priori*.

Consider by way of example Frege's informative identity statement we saw earlier, "Hesperus is Phosphorus." Kripke argues that this statement is metaphysically necessary yet epistemically *a posteriori*. The statement *uses* the names 'Hesperus' and 'Phosphorus', and in so doing it directly refers to those objects. Moreover, names are what Kripke calls "rigid designators," meaning that they refer to the same object in all possible worlds. Then, since the names both refer to the same object the proposition is an identity claim between an object and itself, and so it must be metaphysically necessary. However, it is a contingent fact that those particular names were chosen for that object, and the fact that they were is not something everyone who knows the names will be able to recognize. So the proposition the statement expresses must be *a posteriori* as well. It is a central tenet of contemporary metaphysics that this analysis is right, and moreover that metaphysics proper is concerned with worldly, not representational concepts. I will call this thesis *Textbook Kripkeanism*.

Textbook Kripkeanism is thought to be in tension with hyperintensional theories such as STRUCTURE. The reason is because many of the examples used to motivate hyperintensional theories are rooted in addressing the puzzles of propositional attitude ascriptions. Belief ascriptions are a common example, and take the general form of "*A* believes that *p*" where *A* is some agent and *p* is some proposition. Frege (1892b) famously introduced the puzzling

fact that substituting necessarily equivalent propositions p and q inside of so-called ‘opaque’ contexts, such as propositional attitude ascriptions, may fail to preserve truth. For instance, even if “Audrey believes that Hesperus is Hesperus” is true, it seems as though “Audrey believes that Hesperus is Phosphorus” may still be false, despite Textbook Kripkeanism assuring us that the identity claims are both necessarily equivalent.

Hyperintensional theories of propositional identity may try to account for the failure of substitution to preserve truth in opaque contexts by claiming that the two propositions are in fact not identical despite being necessarily equivalent. However, the argument goes, this is because they are drawing that distinction from the representational differences, the use of two distinct names, between how the propositions are expressed. In fact, this is a plausible interpretation of the motivation for STRUCTURE I argued for in Chapter 3. Recall that my argument claimed that INTENSIONALISM failed to capture a propositional notion of *aboutness*. But isn’t a proposition being about something sensitive to how the proposition is expressed?

According to INTENSIONALISM there is but one necessary proposition, call it p . p is the proposition expressed by “all birds are birds,” “all dogs are dogs,” “ $2 + 2 = 4$,” “there does not exist any integer solutions to $a^n + b^n = c^n$ for $n \geq 3$ when a, b, c are positive,” and many other sentences of natural language. Given that our ways of interfacing with p are for the most part mediated through such sentences, it is easy for us to fail to recognize p is what is being expressed in each of these cases. But this is plainly due to the representational aspects of those mediations. Similarly, the fact that different mathematical statements, each expressing p , have significant differences in the difficulty of proving them and consequently in our ability to recognize their truth is an epistemic, representational consequence of the way they are expressed, not a worldly fact about their truth.

To see another dimension of this attack on STRUCTURE, consider the following two arguments which a proponent of a hyperintensional theory of propositional identity might be tempted to offer against more coarse-grained theories.

Against Extensionalism. Consider the propositions *Audrey is human* (Ha) and *Audrey is fearless* (Fa). Both of these propositions are true on our intended interpretation. As a result, EXTENSIONALISM entails that $Ha = Fa$. However, on our intended interpretation we have that $\Box Ha$ is true while $\Box Fa$ is false. Hence, by **LL** it must be that $Ha \neq Fa$, and so EXTENSIONALISM is false.¹¹

Against Intensionalism. Consider the propositions *Hesperus is Hesperus* (h) and *Phosphorus is Hesperus* (p). Since names are rigid designators, that is, they refer to the same object in all possible worlds, both of these propositions are necessarily true on our intended interpretation. As a result, INTENSIONALISM predicts that $h = p$. But now, let O be an operator which expresses that Audrey believes the proposition O is applied to. Supposing that Audrey has not learned that Hesperus and Phosphorus are two names for Venus, we have that Oh is true while Op is false on our intended interpretation. Hence, by **LL**, it must be that $h \neq p$, and so INTENSIONALISM is false.

Textbook Kripkeanism holds that despite the similarity in form of the two arguments, the first is a sound use of the principles of metaphysics given in *N&N* while the second is not. This is because the concept used to distinguish Ha from Fa in *Against Extensionalism* namely metaphysical necessity, is undisputedly a worldly, metaphysical concept. However, the concept used to distinguish h from p in *Against Intensionalism* is a representational, epistemic concept, since it is sensitive to Audrey's body of knowledge and beliefs.

If the project of higher-order metaphysics is to leverage higher-order languages as a tool of *metaphysics*, so the argument goes, then all of the constructs of our language $\mathcal{L}$ should be metaphysical, worldly constructs and not representational. This is justified by Textbook Kripkeanism, since the worldly concepts are the ones our metaphysical theories should be about. But then the use of representational operators such as O in our language is illicit, and

¹¹Kripke does in fact give an argument of almost this exact form in Lecture III of *N&N*, specifically to argue against identity theories in the philosophy of mind.

so *Against Intensionalism* fails since it cannot even be properly formulated in $\mathcal{L}$ let alone derived in $\mathcal{H} + \text{STRUCTURE}$. Proponents of INTENSIONALISM have thus concluded that “there is therefore a good general reason to doubt that attitude ascriptions can be used to draw any straightforward conclusions about the individuation of propositions,” (Fritz, 2023, p. 51). Our response to the Russell-Myhill paradox, then, should be to keep $\mathcal{H}$ as originally presented and abandon STRUCTURE.

My response to the Textbook Kripkeanism argument is two-fold. First, even if the majority of cases historically used to motivate STRUCTURE involve propositional attitudes, that does not entail that STRUCTURE is in tension with Textbook Kripkeanism principles such as the necessity of identity statements. Whether the tension exists is a separate, logical question about STRUCTURE which needs to be evaluated on its own terms, not through analyzing its motivation. Second, the focus on propositional attitude cases is moreover a non sequitur since metaphysical grounding may be offered as worldly, metaphysical motivation for STRUCTURE which does not conflict with Textbook Kripkeanism. In the rest of this chapter I will develop both of these points in turn.

4.1 Syntactic Structure and the Intended Interpretation

One way of phrasing what the opponents of the structured view think the view gets wrong is that STRUCTURE is too closely tied to the syntax of the statements used to express the propositions. For instance, Williamson claims that “[the structured view] is covertly inspired by *syntax*. The picture is that a declarative sentence expresses a proposition with an overall constituent structure—perhaps that of an inverted branching tree—like that of the syntactic constituent structure of the sentence,” (Williamson, 2024, p. 135). The worry, then, is that identification claims are so fine-grained on the structured view that we will be forced to make distinctions between any two constituents of a proposition which are not represented by exactly the same piece of language. The representational nature of the structured view is then plainly due to the fact that it privileges the syntactic way of representing an object over

its worldly brute identity. My goal now is to show that this understanding of STRUCTURE is false.

Suppose that we go through $\mathcal{L}$ with the intensionalist and throw away any possible predicate which they believe is representational, such as the stipulated O in *Against Intensionalism*. This is something which both the intensionalist and the hyperintensionalist would want to have happen, since both are operating under the higher-order metaphysics framework and hence want $\mathcal{L}$ to capture only worldly concepts. Suppose further that we let the constants ‘ a ’ and ‘ b ’ be names for the same individual, say Audrey.¹² Parallel to the Hesperus and Phosphorus case, we know that $a = b$. Now, if the syntax based objection I outlined above is correct, then there should still be some predicate which holds of a yet does not hold of b since they are syntactically distinct. But then by the contrapositive of STRUCTURE we may conclude that $a \neq b$, contradicting worldly identity. Hence, $\mathcal{H} + \text{STRUCTURE}$ still entails something representational even after throwing away all such operators.

The above argument fails because it makes a conflation between use and mention. When we speak in the language of $\mathcal{L}$, we are *using* the language. Just like in a natural language like English, when we use the name ‘ a ’ we are directly referring to the name’s referent, not to the name itself. To say that there must be something which holds of a and not b because ‘ a ’ and ‘ b ’ are distinct is to conflate the use of the two names in the first case with the mention of the two names in the latter. It will always be the case, after we have thrown out any possible representational pieces of our language, that for any open formula ‘ $\phi(x)$ ’ with free variable ‘ x ’, ‘ $\phi(a)$ ’ and ‘ $\phi(b)$ ’ will have the same truth value when evaluated because the names ‘ a ’ and ‘ b ’ will be used, not mentioned, in those formulas.

One suggestive way of seeing the conflation most clearly can be seen in the following way. Suppose that we stipulatively added a method of quotation into $\mathcal{L}$ and correspondingly added objects of type e which represent each term of the language. For example, for every symbol ‘ s ’ we have its quotation ‘ $\ulcorner s \urcorner$ ’ Recall our statement of STRUCTURE

¹²Beginning now, I will try to practice good quotational hygiene and not make any use-mention conflations myself.

$$(\text{STRUCTURE}) \quad \forall_{\sigma \rightarrow t} F \forall_{\sigma \rightarrow t} G \forall_{\sigma} x \forall_{\sigma} y (Fx =_t Gy \rightarrow (F =_{\sigma \rightarrow t} G \wedge x =_{\sigma} y))$$

and compare it with

$$(\text{SYNTACTIC STRUCTURE}) \quad \forall_{\sigma \rightarrow t} F \forall_{\sigma \rightarrow t} G \forall_{\sigma} x \forall_{\sigma} y (Fx =_t Gy \rightarrow (\ulcorner F \urcorner =_e \ulcorner G \urcorner \wedge \ulcorner x \urcorner =_e \ulcorner y \urcorner))$$

SYNTACTIC STRUCTURE *would* result in such a fine-grained view of propositions that identity conditions would be reduced to syntactic equivalence, and this would surely be representational. However, it is plainly a conflation of use and mention to think that STRUCTURE is equivalent to SYNTACTIC STRUCTURE in the extended language. Moreover, for the reasons I outlined above, it is similarly a conflation of use and mention to believe that adding STRUCTURE to $\mathcal{H}$ would result in as fine-grained a theory of propositional identity as SYNTACTIC STRUCTURE yields in the extended language.

The intensionalist may respond by pointing out that across different models of our language $\mathcal{L}$, ‘ a ’ and ‘ b ’ will quite often be given different denotations. But this means that ‘ $a = b$ ’ will not be a validity of $\mathcal{L}$, and since $\mathcal{H}$ is supposed to be capture exactly those validities, we will not have $\vdash a = b$. But this is true of any two constant symbols $c_{\sigma}^1, c_{\sigma}^2$ for any type σ , and so the only identifications STRUCTURE sees when added to $\mathcal{H}$ are those between the same symbols, since those are the only ones which will turn out to be validities. Hence, STRUCTURE still cuts at the level of syntax.

The problem with this objection is that it is relying on a more general notion of model than the intended interpretation I outlined in Chapter 2. The point is correct when we consider arbitrary set-theoretic formal models of $\mathcal{L}$. But this is true even of something like first-order logic, because it is a general aspect of model theory as a mathematical study of a formal logic. As I argued above, however, we are concerned with an intended interpretation, our actual world and its modal profile, not with validity in an arbitrary class of formal models. This is evidenced by the fact that we are building a *theory* on top of our logic by adding non-logical axioms such as STRUCTURE in order to carve out the intended interpretation.

To see this point more clearly, consider an analogy with mathematical logic and arith-

metic. The goal of arithmetic is to study a particular structure, the standard model of arithmetic $\mathbb{N}$. Specifically, we want to understand the set of all first-order sentences in the language of arithmetic which are true of that model, which we call $\text{Th}(\mathbb{N})$. The general way that we may do this is to list off a specific axiomatization of $\text{Th}(\mathbb{N})$, that is, a list of axioms all of whose models are isomorphic to $\mathbb{N}$. It then follows from the soundness and completeness theorems for first-order logic that the sentences ϕ provable from that axiomatization are exactly the sentences in $\text{Th}(\mathbb{N})$. Now, in the case of arithmetic, Gödel’s incompleteness theorems tell us that no such axiomatization meeting certain ‘usability’ conditions exists. But the strategy is generalizable and works in other domains: if you want to understand a particular structure aim to find a set of axioms whose models are all isomorphic to that structure. If the logic is sound and complete with respect to that class of models, then the provable statements from those axioms will be exactly the statements true of the structure.

Relating the arithmetic example back to the project of higher-order metaphysics, we may say that our goal is to propose further axioms to $\mathcal{H}$ which narrow down the possible space of interpretations until we land directly on the intended interpretation. That is what it means to develop a higher-order theory of propositional identity. So even if we were to step back from the intended interpretation and engage in a set-theoretic model theoretic study of $\mathcal{L}$, we would still not be interested in general validity across all models, but instead just across the models isomorphic to the formal surrogate of the intended interpretation. That is, would then only be concerned with models which fix the metaphysical facts.

In summary, the claim that STRUCTURE carves at the level of syntax and hence introduces representational facts into our worldly logic $\mathcal{H}$ is false. The goal of adopting a theory of propositional identity is to pin down the metaphysical facts of our world, those making up the intended interpretation of our language, and the fact that we *use* that language means that we do not conflate syntactic representation with worldly individuals. I conclude, then, that STRUCTURE is not in tension with Textbook Kripkeanism.

Before concluding this section, I wish to consider one last objection, namely that my

argument made heavy reliance on $\mathcal{L}$ having an intended interpretation, one which fixed the meaning of every term of the language. But why should we be so sure that such an interpretation exists? Although intensionalists such as Fritz (2023) and Williamson (2024) both agree that there is such an intended interpretation, others may object that we are asking too much of our language.

To answer this question, it is important to be clear about how we are using the word ‘interpretation’. As I alluded to above, in the general setting of mathematical logic an interpretation of a language is a function which assigns values (and n -tuples of values) to the non-logical vocabulary of the language in addition to assigning truth-values to its sentences. But on the standard understanding of what a function is, such an interpretation will be set-sized. So, if we are using ‘interpretation’ in this standard way, one would be warranted to have skepticism that our intended interpretation fixes every metaphysical fact since there are just too many objects, let alone higher-order entities, for such an interpretation to be set-sized. In other words, there wouldn’t be such an interpretation to be found. But this is not the way I have been using ‘interpretation’. As I argued above, we are not concerned with the formal, set-theoretic model theory of $\mathcal{L}$. Just as natural languages do not depend on a set-theoretic semantics to give their sentences meaning, our higher-order language does not depend on a set-sized interpretation function to give it meaning. Both kinds of languages benefit greatly from the existence of set-theoretic semantics which allows a scientific study of many of their properties. But, clearly, that semantics is not a necessary condition on their intelligibility or use. So, objections based on cardinality considerations are unwarranted. We can be sure that there is an intended interpretation because we are sure that there is a metaphysical reality.

4.2 What Motivates Structure?

Now, it is not clear to me why proponents of INTENSIONALISM focus on attacking motivations for hyperintensional theories based on propositional attitude reports. As my arguments in

the previous section highlighted, the alleged issues STRUCTURE was supposed to inherit from those motivations do not exist and moreover that conclusion was reached in an environment which explicitly eschewed such opaque operators. Still, I will grant for the sake of argument that, since STRUCTURE is being proposed as a *theory*, it would be copacetic with the goals and methodology of higher-order metaphysics to have a soundly worldly motivation. And, after all, it is true that historically propositional attitude cases were the first significant cases to be discussed and they remain important for issues at the intersection of metaphysics and the philosophy of language. So we owe it to the intensionalist to provide a worldly motivation for STRUCTURE. In this section, I offer metaphysical grounding as the best candidate.

Contemporary work on metaphysical grounding owes much to the work of Kit Fine and Jonathan Schaffer, among many others. Schaffer (2009) contrasts the Quinean approach to metaphysics that focuses on the question “what there is” with a neo-Aristotelian approach that asks “what grounds what.” For the Quinean, questions about ontology come down to what entities are contained in the domain of quantification of our best theories of reality. There is, in general, no distinction between any of those entities being primary or more fundamental than any other. The neo-Aristotelian view, on the other hand, takes it at face value that most thing we speak of do exist in an appropriate sense, and the fundamental question of ontology is thus to understand the *grounding* relationships between them. On this view, the existence and nature of some entities are grounded in the existence and nature of others. This grounding, moreover, is not a causal sort of relationship. It is a deeper relationship that aims to track the fundamental constituency of the entities themselves, and not just what, possibly contingent, facts lead to their being. For the metaphysician who believes in ground, answers to questions such as “do numbers exist?” are a resounding “of course.” The interesting question, they argue, is “*how* do numbers exist?”

Proponents of ground face an immediate question: what sorts of entities stand in the grounding relation? Phrased another way, what sorts of entities may ground other sorts of entities? There are two popular theories which answer this question in different ways:

entity-grounding and propositional-grounding (Bliss, 2024, pp. 4–6). Entity based theories of grounding allow the relation to hold between objects, those of type e in our HOL framework. So, a particular table may be grounded in its legs and top surface, an album may be grounded in its songs and artist, a government may be grounded in its constituent branches and officials, and so on. In contrast, propositional-grounding holds that propositions, entities of type t , are what stand in the grounding relation. Hence, the proposition *the table exists* is grounded in the propositions *the legs of the table exist* and *the surface of the table exists*. I will be adopting the latter view going forward.

Prima facie, our higher-order language $\mathcal{L}$ is a perfect vehicle for exploring the nature of metaphysical grounding. Formalizing the grounding relation in a first-order logic requires exactly the kind of restricted quantifications over abstracta which higher-order metaphysicians think should be avoided. Instead, we may leverage the power of $\mathcal{L}$ by introducing directly a new non-logical constant $\prec^{t \rightarrow t \rightarrow t}$ as a higher-order grounding relation between propositions. In the spirit of higher-order metaphysics, we may then investigate what sorts of axioms we should adopt so that the use of $\prec$ in our language correctly captures the nature of metaphysical grounding. But it is well known that the vast majority of theories of grounding which have been argued for lead to hyperintensional distinctions between necessarily equivalent propositions (Schaffer (2009); Fine (2012)). Hence, metaphysical grounding gives us a non-representational motivation for exploring hyperintensional theories of identification such as STRUCTURE.

Fritz anticipates this sort of a response and argues that it has a fundamental flaw: plausible and generally agreed upon principles which govern the grounding relation also generate a form of the Russell-Myhill paradox. Consider the following two principles:

$$(\wedge \prec) \quad \forall p \forall q \forall r (p \prec (q \wedge r) \leftrightarrow (p = q) \vee (p = r))$$

$$(\forall \prec) \quad \forall p ((p \prec \forall q \phi(q)) \leftrightarrow \exists q (p = \phi(q)))$$

As argued for by Fine (2012), these are very plausible accounts of *partial* grounding: $\wedge \prec$ says that conjunctions are grounded in their conjuncts and $\forall \prec$ says that universal quantifications

are grounded in their instances. Yet Fritz shows that these two principles entail certain cases of STRUCTURE and hence are also inconsistent with $\mathcal{H}$. From this, he draws the conclusion that grounding relations must themselves be representational and not worldly.

I believe that Fritz is making the mistake of putting the cart before the horse here. It is worth considering his own way of phrasing the objection. First, consider what he says about Russell-Myhill and STRUCTURE:

The Russell-Myhill argument shows that propositions cannot be structured. This provides a first illustration of how the method of improving upon formal talk of propositions and modalities using $\mathcal{L}$ bears fruit. Using basic logical principles [such as β -equivalence], non-obvious consequences can be derived. It would be a mistake to regard this as a sign that something has gone wrong. Recall that these basic principles are adopted stipulatively, as part of the metasemantics of $\mathcal{L}$. Assuming the introduction of $\mathcal{L}$ succeeds, the truth of these principles is therefore not in question. — (Fritz, 2023, p. 55)

It is clear from this quote that Fritz takes the axioms of $\mathcal{L}$ to be beyond reproach. At the same time that he declares them as being adopted stipulatively, he also claims that any contradiction they help generate, such as Russell-Myhill, is evidence that the other principles involved in the derivation are at fault, not those of $\mathcal{H}$.

Now consider what Fritz has to say about the fact that the above principles of partial grounding derive instances of STRUCTURE and hence also Russell-Myhill:

We might sum this observation up as showing that central grounding principles require propositions to be finely individuated to an inconsistent degree. ... Little progress can be expected from using our intuitions to argue for and against the view that grounding is a relation of metaphysics, as opposed to tracking features of our representation of reality. By focusing on the individuation of propositions, and the demands of grounding on such individuation, it is possible to give a general argument that grounding must be of the latter, representational, nature, and

so should not be added to our metaphysical vocabulary. This in turn has consequences for our views about propositional individuation, since it removes one further potential reason for thinking that propositions are individuated relatively finely. — (Fritz, 2023, pp. 61–62)

Fritz alludes to the fact that there is disagreement in the literature about whether metaphysical grounding is a worldly or representational concept. But, Fritz holds, the inconsistency of the two principles of partial grounding brings us to the conclusion that metaphysical grounding must be of a representational nature without needing to rely on competing intuitions. Instead, we can just look at the logical facts: the principles are inconsistent with $\mathcal{H}$. Hence, this is just another case of higher-order metaphysics bringing clarity to a muddled debate in metaphysics and providing a decisive answer. Unfortunately for Fritz, this argument is entirely circular.

The ‘general argument’ against metaphysical grounding Fritz alludes to above seems to me to be the following:

- (1) Any concept whose logical principles derive the Russell-Myhill paradox when added to $\mathcal{H}$ is representational.
- (2) The metaphysical grounding relation derives the Russell-Myhill paradox when added to $\mathcal{H}$.
- (3) Therefore, metaphysical grounding is representational.

But what is the justification for (1)? According to the two quotes above, for Fritz (1) holds because any principle which derives the Russell-Myhill paradox must be hyperintensional enough to derive STRUCTURE, and the motivation for adopting STRUCTURE comes from representational distinctions like those made in propositional attitude ascriptions. But this justification for (1) clearly begs the question. If the representational nature of STRUCTURE is supposed to be derived from its motivation, and metaphysical grounding is offered as

a new motivation for adopting STRUCTURE, it is circular to then object to metaphysical grounding on the grounds that it leads to STRUCTURE. But then what is the justification that STRUCTURE is representational? Because its motivation is metaphysical ground? We've run in a circle.

What the Russell-Myhill shows is that $\mathcal{H} + \text{STRUCTURE}$ is inconsistent. What it does not show is that, therefore, STRUCTURE is a representational principle. The paradox itself is silent on which principle is the 'offender' in the sense that the paradox itself does not entail anything further than the existence of a contradictory pair. In fact, the spirit of exploration of new principles of metaphysics which all proponents of higher-order metaphysics champion compels us to explore both alternatives, since, as Fritz says, the principles of $\mathcal{H}$ were stipulative. Having given what I believe to be a thorough defense of STRUCTURE as a worldly principle of propositional identity, I will next consider arguments against β -equivalence and propose a revision to $\mathcal{H}$ which is consistent with hyperintensional principles such as STRUCTURE.

5 Revising Our Higher-Order Logic

In the last section, I quote Fritz as saying that the principles of $\mathcal{H}$ were stipulative. This might feel in tension with the matter-of-fact style in which they were introduced in Chapter 2. Although I did highlight the fact that COMPREHENSION is derivable using β -equivalence and mention its relevancy for the arguments I am about to give, I did not spend any time discussing the motivation for β -equivalence or what potential alternatives were available. It is now time to consider these questions.

Recall that the historical introduction of the λ -calculus by Church came in the form of a model of computation. The principle of β -reduction, moving from an application $(\lambda x.Fx)a$ to its reduced form Fa through substitution, is the heart of the model of computation. It is supposed to capture the informal notion of an effective procedure, one which is local, determinate, and finite. Considering our higher-order logic $\mathcal{H}$, this idea would seem to justify only the left-to-right implication that $(\lambda x.Fx)a \supset Fa$. But what about the other direction, β -expansion? The conception of the λ -calculus as a model of computation does not justify β -expansion in the same way. This is because β -expansion is in general a non-deterministic procedure. Recall that from Fa , we may β -expand to at the very least either $(\lambda x.Fx)a$ or $(\lambda X.Xa)F$. But from a computational standpoint, this indeterminism is not acceptable, since computations unfold over time and have an unambiguous history. The justification, then, of adding β -expansion to $\mathcal{H}$ is not derived from the conception of the λ -calculus as a model of computation. So, what is its justification?

Dorr offers an argument based off of a translation scheme from natural language: “An object x is such that it is rectangular and it is equilateral if and only if x is rectangular and x is equilateral. More generally: x is such that ... it ... iff ... x ...,” (Dorr, 2016, p. 52). The idea is that phrasing the predication as “ x is such that it is rectangular and it is equilateral” corresponds to an application $(\lambda z.Rz \wedge Ez)x$ while phrasing a predication as “ x is rectangular and x is equilateral” corresponds to the simple conjunction $Rx \wedge Ex$. But, so the thought goes, these two ways of phrasing the predication, if not identical, are at least

equivalent, and hence we should have at least our weaker scheme of extensional β

$$\mathbf{E}\beta \vdash (\lambda\nu.\psi)\alpha \leftrightarrow \psi[\alpha/\nu]$$

But, Dorr continues, since we are working strictly within worldly, non-opaque contexts, there shouldn't be any problem with replacing one of the predications with the other even if they happen inside of some other context. So, we may generalize this to full β -equivalence

$$\beta\text{-}\mathbf{Eq} \vdash \phi[(\lambda\nu.\psi)\alpha] \leftrightarrow \phi[\psi[\alpha/\nu]]$$

Is this a good justification for accepting β -equivalence? Its main premises are that there is no distinction between the two ways of expressing the predication in natural language and that we are working in a higher-order language with no opaque contexts. As I argued in the last chapter, even the hyperintensionalist agrees with the latter point. But it isn't clear that the first premise is correct. In fact, as Dorr points out, proponents of metaphysical grounding such as Gideon Rosen and Kit Fine have shown that standardly accepted principles of the grounding relation conflict with β -equivalence (Dorr, 2016, pp. 52-55).

The derivation goes as follows. Because Fa grounds $(\lambda x.Fx)a$, and no proposition grounds itself, we must have $Fa \neq_t (\lambda x.Fx)a$. Now, it follows from an instance of β -conversion that $(Fa =_t (\lambda x.Fx)a) \leftrightarrow (Fa =_t Fa)$. But since by **Ref** we do have $(Fa =_t Fa)$, and by **MP** we have $Fa =_t (\lambda x.Fx)a$, contradiction. This is a much more direct way in which principles of metaphysical grounding conflict with β -equivalence than by simply deriving **STRUCTURE** and hence Russell-Myhill.

Now, Dorr does have a response, and in particular he denies the premise that on a worldly conception of grounding Fa grounds $(\lambda x.Fx)a$. What the proponent of metaphysical grounding should do from here is an interesting question, but I will not pursue it further. It is dialectically important for the hyperintensionalist to defend ground from this objection, since it means ground both supports the structured view and independently targets β -equivalence. But I believe there is another important way of undermining $\mathcal{H}$ which is more pressing. In particular, I believe that the full **COMPREHENSION** scheme is something

which a higher-order *logic* such as $\mathcal{H}$ should not contain. And since, as we showed in Chapter 2, β -equivalence allows us to derive every instance of COMPREHENSION, any argument undermining COMPREHENSION is a fortiori an argument against β -equivalence.

5.1 Higher-Order Comprehension Is Too Strong

My argument against COMPREHENSION proceeds on two fronts. What both of these fronts have in common is that they take seriously the claim that $\mathcal{H}$ is supposed to be a *logic*. One central philosophical criterion that has been placed on logicity over the past century of analytic philosophy has been that logic is as neutral as possible with respect to the way things may be. Whether or not, say, “Hesperus is Phosphorus” is *logically* indeterminant, despite being *metaphysically* necessary, since it depends on the referents assigned to the names. What I will show here is that COMPREHENSION, and thus β -equivalence, makes too strong of a commitment to the nature of metaphysical reality to be part of our logic.

My first objection to COMPREHENSION comes from what I take to be an intuitive response to the derivation of the Russell-Myhill paradox. Recall that the version of Cantor’s theorem we prove may be roughly paraphrased as saying that no higher-order operator on sentential operators may be injective on its arguments. What this means is this. For any higher-order operator f of type $(t \rightarrow t) \rightarrow t$, there exists at least one pair O, M of operators of type $t \rightarrow t$ such that $O \neq_{t \rightarrow t} M$, yet $fO =_t fM$. But now take as an example any contingently true proposition p (say, that you are reading this thesis right now). Consider as an example $f = \lambda N. Np$. f takes an operator, such as $\neg$ or $\Box$, and applies it to p . What Cantor’s theorem tells us is that there must exist at least two distinct operators O, M such that $Op =_t Mp$. But is this really a truth of *logic*?

Another way of expressing what I take to be wrong is that COMPREHENSION *limits* the possible interpretations of $\mathcal{H}$ in such a way that ensures that for each higher-order operator f , there *must exist* within the interpretation at least a pair of operators with the above property. We cannot have any other such interpretation, since the interpretation needs to

make COMPREHENSION valid. But is the lack of any injective operator really a truth of logic, a neutral principle with respect to the way things may be? Why shouldn't it be at the very least possible that there does exist such an injective operator, especially on a proposition such as p whose truth value is not even fixed across all such models?¹³

The standard response to this is that the existence of such an injection contradicts logic in the same manner as the existence of the Russell set does. As the derivation of the Russell-Myhill paradox shows, the argument is a diagonalization. There are just too many propositions for there to be such an injective function. But the claim that there *are* this many propositions is a consequence of COMPREHENSION. The picture of the totality of propositions which *Comprehension* commits us to is one which is completely given and as a result must void the possibility of certain fine-grained distinctions as a result, for otherwise there would just be too many propositions and we would get a Cantorian contradiction. However, this is not the only consistent conception of the totality of propositions.

One alternative conception is the ramified hierarchy picture, which was deployed by Russell to deal with the set theoretic paradoxes. On the ramified hierarchy picture, each proposition has a level which is one higher than the level of its constituents. On the ramified view, each level contains higher-order 'copies' of the entities below it, and thus Russell posited the axiom of reducibility which allows moving between those higher-order copies to their lower-order counterparts. Although this conception of the totality of propositions does come as completely given, just like the COMPREHENSION conception, as Russell established, the ramification is enough to block the derivation of the paradox. Now, defenders of β -equivalence such as Dorr (2016) and hyperintensionalists such as Linnebo (2024) alike all agree that the ramified hierarchy picture is at best ugly and worst totally implausible view. But it is still a consistent conception of the totality of propositions.

Another, more plausible alternative is that presented by Linnebo (2024). On this picture,

¹³The argument I am giving here is very reminiscent of the one given in Kaplan (1995), which introduced a similarly compelling 'propositional paradox' into the literature, one which threatens the standard understanding of possible world semantics. The upshot for Kaplan is that "the metaphysical should not be confused with the logical," the same point I am here trying to stress.

which he calls the hierarchial picture, the totality of propositions is merely a potential totality, not a completed one. The hierarchial levels of the totality proceed upwards through ontological dependence on the levels below them. It follows then that the resources available at any given level are only those below it. As Linnebo says, “it is hard to understand how an unbounded universal generalization [such as the instance of COMPREHENSION which is used to generate the diagonalization predicate in the Russell-Myhill argument] can be grounded at some particular stage of the hierarchy, given that most of its instances will not be available at that stage,” (Linnebo, 2024, pp. 215-216). Thus, this hierarchial view also avoids the Russell-Myhill paradox, but it does so by denying the COMPREHENSION view that there is a completed totality of propositions.

I take it to be consonant with the neutrality of logic that it should not decide one way or the other what the totality of propositions must look like, just as the logic itself is not strong enough to decide all cases of propositional identity. Surely, I am not trying to prohibit the possibility of proposing the COMPREHENSION picture of the totality of propositions as a *theory*. But I am taking the position that this does amount to a theory choice, and not something which we should take to be a general principle of logic proper.

My second argument against COMPREHENSION is that it implicitly adjudicates the debate between sparse and abundant conceptions of properties. This distinction between properties, first crystallized by David Armstrong and David Lewis (see Orilia and Paoletti (2025)), is again a contemporary refinement of an old idea. A sparse conception of properties says that there are certain, yet very few, joint-carving properties which are in a strong sense natural and fundamental. From these we can construct many other sorts of properties to aid in the semantic act of predication, but these are in a strong sense derivative of natural properties. The abundant view makes no such distinction, and instead holds that there are as many properties as there are ways of doing such predications, none more fundamental in than others, or at least not in the way that the sparse view holds them to be.

Now, given that the sparse-abundant distinction is a central debate in contemporary

metaphysics, shouldn't the *logic* of higher-order metaphysics remain neutral and act as an arena for the opposing theories to contest each other? Consider these passages from Williamson and Fritz.

In this book, the central application of higher-order modal logic is not to natural language semantics, but to logic and metaphysics. Is a restriction to natural properties and relations motivated metaphysically rather than semantically? ... Logic restricted to natural properties and relations is pathetically weak. ... Metaphysics based on weak logic wastes its time taking crank theories seriously. It needs a strong logic with laws of unrestricted generality. For example, [the correct] version of Leibniz's Law says that identicals have *all* their properties (however unnatural) in common. [Footnote: In the terminology of Lewis (1983), a sparse theory of universals such as that defended by Armstrong (1978) is no good for our purposes.] — (Williamson, 2013, pp. 226-227)

Since Lewis (1983) it has become common to distinguish between two conceptions of properties, a *sparse* and an *abundant* conception. ... In the time since Lewis's influential paper, it has become common to use talk of properties in the abundant sense, and to capture the sparse conception of properties by distinguishing certain properties as *natural*. This will be the approach taken in what follows: properties will be taken to be abundant, and the question whether there is a role to play for some of them to be distinguished as natural will be set aside for another occasion. — (Fritz, 2023, p. 16)

It is clear from both of these passages that Williams and Fritz both agree that, despite it being a live question of what the right conception of properties is, our higher-order logic should adjudicate on the side of abundance if only for the pragmatic reason that existentially and universally generalizing on any open formula $\phi(x)$ is a useful thing for us to do. And of course, COMPREHENSION is exactly the way in which $\mathcal{H}$ does adjudicate this way. For

any predication $\phi(x)$ for a free variable x , COMPREHENSION assures us that there exists a higher-order property to be the value of that predication. So, once again, COMPREHENSION is failing to capture the neutrality inherent in being part of an underlying logic.

Is this lack of neutrality avoidable while still retaining COMPREHENSION? As Fritz alludes to, although it is customary to set up our languages in an abundant way, we may introduce new operators into our language which capture the naturalness inherent in the sparse conception. The problem with this move, however, is that hyperintensional concepts like metaphysical grounding are exactly those notions deployed to capture naturalness and fundamentality. Given that COMPREHENSION and metaphysical grounding are also mutually inconsistent, there is no room for a sparse conception of properties in $\mathcal{H}$ and once again the intensionalist runs the risk of question begging against the opposing side in the debate.

I conclude from these two arguments that COMPREHENSION, and thus β -equivalence, are just too strong of principles to be included in our higher-order logic. They commit us to controversial views in metaphysics which require a theory, not a logic, to adjudicate for.

5.2 A Revised Logic

Here, finally, is the presentation of my alternative logic, which I shall call $\mathcal{K}$. Like $\mathcal{H}$, $\mathcal{K}$ is a subset of the sentences of $\mathcal{L}$ which we take to be the logical truths of the language. Moreover, as will be evident by how I present it, $\mathcal{K}$ is in fact a proper subset of $\mathcal{H}$.

Formally, let $\mathcal{K}$ be the smallest subset of $\mathcal{L}$ which is closed under the following axioms

PC $\vdash \phi$ when ϕ is a tautology from classical propositional logic

UI $\vdash \forall_\sigma \xi \supset \xi\alpha$

EG $\vdash \xi\alpha \supset \exists_\sigma \xi$

Ref $\vdash \alpha =_\sigma \alpha$

LL $\vdash \alpha =_\sigma \beta \supset (\xi\alpha \supset \xi\beta)$

$\beta\text{-Re} \vdash \phi[(\lambda\nu.\psi)\alpha] \supset \phi[\psi[\alpha/\nu]]$

and the following rules of inference

MP If $\vdash \phi$ and $\vdash \phi \supset \psi$, then $\vdash \psi$.

Gen If $\vdash \phi \supset \psi$ and ν does not occur free in ϕ , then $\vdash \phi \supset \forall_\sigma \nu \psi$

Ins If $\vdash \phi \supset \psi$ and ν does not occur free in ψ , then $\vdash \exists_\sigma \nu \phi \supset \psi$

Nec If $\vdash \phi$ then $\vdash \Box\phi$.

It is immediately noticable that, perhaps somewhat anticlimactically, we have just replaced β -equivalence with β -reduction. But I hope it is clear from everything I have argued over the last three chapters that this is a non-trivial decision.

I will end this section by considering one last objection. Why not just drop the β -related axioms altogether? After all, taking our logic to be $\mathcal{K}$ leaves open the possibility of adopting some form of β -expansion as a *theory* on top of $\mathcal{K}$ provided that we have an appropriate argument for the metaphysical truth of that principle. Why shouldn't this same neutrality apply to β -reduction? Moreover, if the response is because β -reduction somehow is neutral with respect to the field of possible theories while β -equivalence is not, the following seems to be a counterexample. Recall the *de dicto/de re* distinction with respect to propositional operator, say O . The best way of expressing that distinction in a language with λ -abstraction is as the following:¹⁴

$$De\ Dicto \quad O((\lambda x.Fx)a)$$

$$De\ Re \quad (\lambda x.O(Fx))a$$

But if our logic contains β -reduction, then this distinction will in a significant sense collapse,

¹⁴See Fitting and Mendelsohn (2023) Chapters 14-16 and 19 for an extended discussion on this point and how this particular scope distinction, as argued by Kripke, is a generalization of the *de dicto/de re* distinction as well as the rigid/non-rigid distinction and Donnellan's attributive/referential distinction.

since each of these entails $O(Fa)$.¹⁵ The claim that there is no such strong scope distinction with respect to modal operators is itself a metaphysical claim, and so we should adopt a background logic which is neutral to it.

My response to the above argument is two-fold. First, as I mentioned at the start of this chapter, β -reduction does have a particularly nice justification stemming from the computational interpretation of the λ -calculus. I believe that interpretation maps onto the concept of predication in the correct way. Recall that in the presentation of the *sui generis* interpretation of $\mathcal{L}$ in Chapter 2, I mentioned that the adverbial paraphrase account was motivated by the idea that higher-order generalizations describe the ways in which objects are, not what there is. It makes sense then that an application such as $(\lambda x.Fx)a$ will reduce to Fa , since it eventually should bottom out in an ordinary claim about objects.

Secondly, that β -reduction in some way collapses the scope distinction is not actually in tension with Textbook Kripkeanism. Although this is not the original way in which Kripke defined them, an equivalent way of informally defining rigid designators is as terms in our language for which the above scope distinctions do not make a difference, that is, just when both sentences are materially equivalent (Fitting and Mendelsohn, 2023, p. 299). This would then only be a problem if $\mathcal{L}$ contained non rigid-designators, which Textbook Kripkeanism tells us it should not.

It is true that if we wanted to expand our language with some sort of non-rigid formalism, such as a definition description operator ι or an agent index knowledge operator K_a , we would need to either stipulate particular exceptions for β -reduction within their scope, or else drop β -reduction entirely. This is often done in Montague style higher-order logics commonly used in formal semantics and the philosophy of language. But this is clearly introducing representational aspects into our language, and even the proponents of β -equivalence would agree that something in the underlying logic must then break down. It is thus not an objection to taking $\mathcal{K}$ to be the logic of $\mathcal{L}$, which contains only worldly concepts by

¹⁵This is not as bad as each of the two forms implying the other, which would happen under full β -equivalence. However, see the response below.

stipulation.

6 Conclusion

I will conclude this thesis with a brief summary of what I take to have happened over the last five chapters. Then, I will stress what I take to be the two substantive points to take away, one local to the project of higher-order metaphysics and one more broadly of philosophical methodology. Finally, I will say a few words on the dialectical position I think higher-order intensionalists find themselves in.

We began by presenting higher-order logic as a new vehicle for exploring complicated issues in contemporary analytic metaphysics. We applied the increased expressivity of the logic to bring new clarity to the old problem in analytic metaphysics of the identity conditions of propositions. It turned out that a familiar view, that propositions are individuated according to their structure, is inconsistent when added to our logic. We were then faced with a decision: do we abandon the structured view of propositions or do we change our underlying logic? We took the latter route, first by showing that common arguments against choosing structure fail and then by showing that we have independent reason to change the underlying logic to be more compatible with other concepts in contemporary metaphysics. What we ended up with was a weaker logic which allows for a far wider range of theories to be built up and compared.

What value do I believe there to be in all of this? First, I do take there to be local value in these arguments for the higher-order metaphysics project. Grounding, essence, fundamentality, and so on are the contemporary manifestation of very old ideas in metaphysics, and the fact that they have stuck around for this long and admit of formalizations means that they deserve to be studied as thoroughly as possible. I hope that at least some of these arguments may sway some higher-order metaphysicians to believe that adapting our background theories to accommodate these concepts is a necessary part of the project.

What I take to be the broader value in this thesis, however, is a general methodological point already noticed by Kripke (1976). In the early 1970s, three papers were published by logicians who claimed to show as a ‘hard logical fact’ that substitutional quantification

must reduce to objectual quantification. In a genuine tour de force of a paper, Kripke shows that the claims in the three papers are mathematically false and develops what is now the standard theory of substitutional quantification in a mathematically consistent and philosophically coherent way.

At the end of the paper, Kripke gives a “methodological sermon” wherein he decries the brute use of mathematical theorems (nevermind falsehoods) to “grind out” philosophical results (Kripke, 1976, pp. 407-416). Among the methodological points he stresses are “philosophers should not confuse their own particular philosophical doctrines with the basic results and procedures of mathematical logic,” “the tendency to propose technical criteria with the aim of excluding approaches one dislikes should be curtailed,” and “unless a technical criterion has been given a rigorous philosophical justification which really justifies us in maintaining it *a priori*, I suggest that if an author wishes to work under its constraints, he should regard it as a hypothesis within which he chooses to work.” It seems to me that the attitude expressed by the claim that the Russell-Myhill paradox definitively proves, by a purely logical argument, that metaphysical reality is not structured offends on each of these methodological principles.

I will end this thesis with a few words about the higher-order metaphysics project itself. It is clear, given all that I have said about metaphysical grounding, that the recent turn towards hyperintensional distinctions in worldly metaphysics does not play nicely with higher-order logic.¹⁶ One reaction to this, the one I pursued in this paper, is to accept the higher-order metaphysics picture and find a plausible way to recover consistent notions of grounding and related concepts. But as is clear from my overall goal in this thesis, that requires making changes to the underlying logic. Opponents of hyperintensionalism have referred to these changes as *ad hoc* and the result of bad theory building via overfitting of data (see especially Williamson (2024) on this point). I resist the claim that the changes are *ad hoc*, since they have the neutrality of logic as an independent motivation. But still, perhaps the alternative

¹⁶This way of phrasing it was expressed to me by Mark Jago, one of the main theorists, along with Kit Fine, of hyperintensional truthmaker semantics, at a conference on formal methods in contemporary metaphysics.

strategy of abandoning higher-order logic and its *sui generis* interpretation is better, since at the very least we have a much clearer picture of what it is we are talking about in the first-order realm. If the higher-order metaphysicians want to keep the hyperintensionalist theorist within their own arena, they need to do much more to explain what exactly the intended interpretation of our higher-order language is and why it provides a particularly perspicuous grasp on the fundamental nature of reality which theorizing in first-order languages does not. Otherwise, the hyperintensionalist's response to the "purely logical arguments" that their theories are proven false in higher-order frameworks will be a justified, "so what?"

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